

Original covariance growth and the measured arithmetic resolvent

Split-Zero research programme

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The original invariant covariance can now be approximated to a prescribed accuracy with a total of $O(kq + k \log(1/\eta))$ scalar samples. The proof retains every invariant row and every lower-root coefficient, and gives finite bounds for errors in both. Comparing the low and high cutoffs directly also controls the full complementary minimum when a small row subspace is separated.

The measured arithmetic resolvent has an exact matrix correction from its observation kernel. This paper reconstructs that correction, determines its smallest contributing kernel space and bounds the effect of coherent source errors by its actual nonzero ranks. All complex cross blocks are retained. The incoming signed determinant calculation is reproduced with its stated analytic inputs and original four-cutoff signs.

The four-label ES receiver has a separate complete calculation: all four first singular corrections, its inverse-column correction, the exact exceptional kernel and cokernel, and every fixed scalar observation of the slow heat. Its original arithmetic scope is preserved.

The period-dependent native kernel coefficient through kq and the separate complex-current signs remain unevaluated. The results here supply proved reductions, finite error bounds and further exact values; they do not assign those remaining quantities. Complete inherited proof sources, their precise reading records and human citations accompany this edition.

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The actual minimum and the kernel correction

The onto observation $\Lambda : E \rightarrow B$ has its original kernel K . The source metric G determines the minimum section L and value metric Q . The exact splitting map and energy are

$$U_0(b, a) = Lb + I_K a, \quad U_0^* G U_0 = \begin{pmatrix} Q & 0 \\ 0 & H_K \end{pmatrix}, \quad U_0^* T^* G T U_0 = \begin{pmatrix} E_{BB} & X^* \\ X & E_{KK} \end{pmatrix}.$$

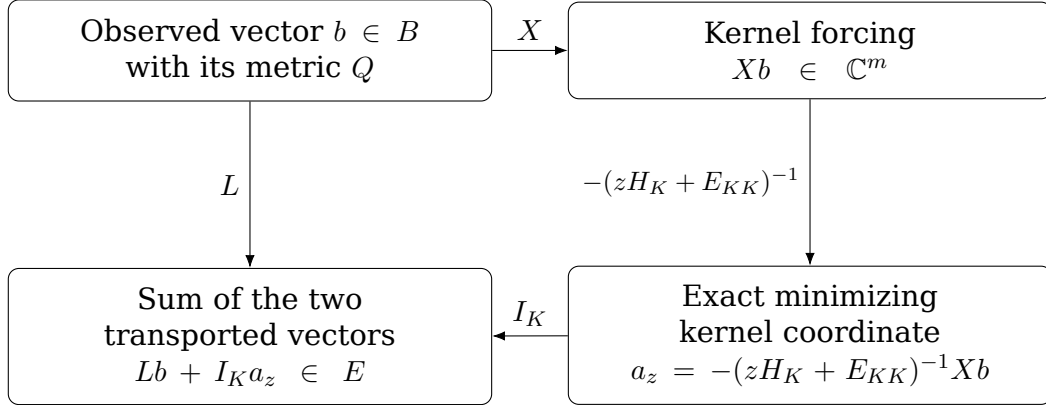


Figure 1: The exact full minimum in RM1-8. The arrows labelled L and I_K supply the two summands of the full vector. Both cross blocks remain. The kernel correction is $D(z) = X^*(zH_K + E_{KK})^{-1}X$. The correction subtracts from $zQ + E_{BB}$, and the observed endomorphism is $\mathcal{Y}(z) = z[zQ + E_{BB} - D(z)]^{-1}Q$. This is the finite Feshbach-Schur construction; the proof cites Dusson, Sigal and Stamm at its precise original-author source location.

The contributing kernel space is the span of $A^j C b$, where $A = H_K^{-1/2} E_{KK} H_K^{-1/2}$ and $C = H_K^{-1/2} X Q^{-1/2}$. Its exact dimension is the rank of the phase-bearing moment matrix $[C^* A^{i+j} C]_{i,j=0}^{m-1}$ (RM12-14). Reducing kernel directions with zero coupling have an explicitly proved cancellation; none is discarded on the basis of a rank count alone.

An exact finite comparison of the three actions

RM22 evaluates the example $G = I_3$, $T = \text{diag}(0, 2, 3i)$ and $\Lambda = \begin{pmatrix} -1 & 1 & 0 \\ -i & 0 & 1 \end{pmatrix}$. Its kernel frame is $(1, 1, i)^\top$ and

$$D(z) = \frac{1}{9(3z + 13)} \begin{pmatrix} 1 & 14i \\ -14i & 196 \end{pmatrix}.$$

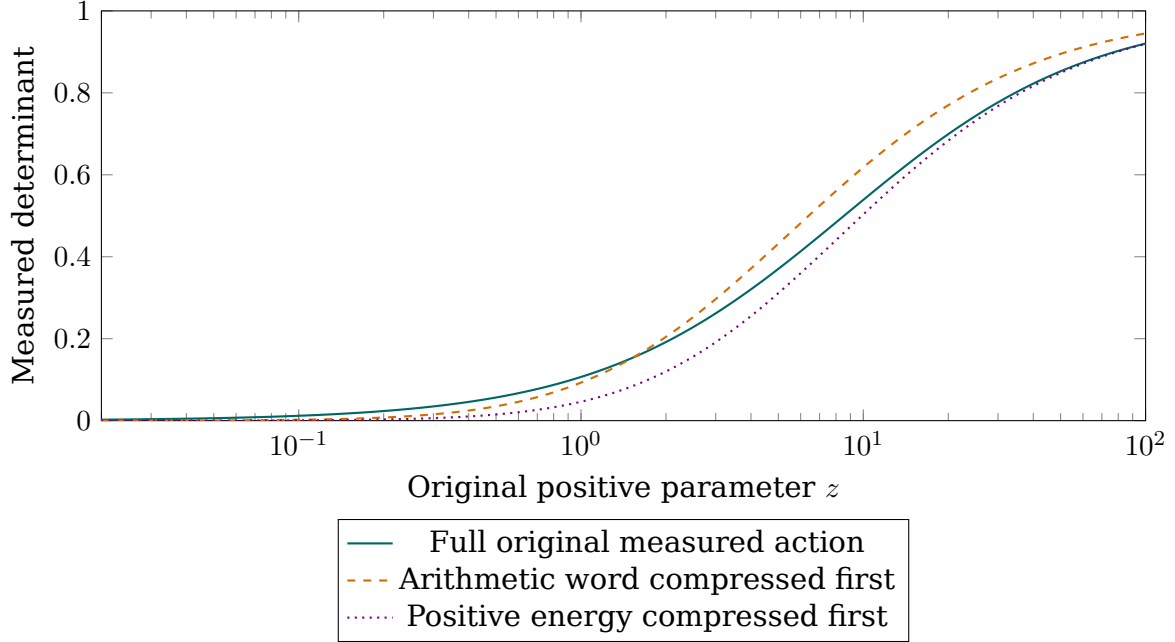


Figure 2: Exact rational functions from RM22, sampled only to draw the curves. The full action is always above the compressed positive-energy action (RM15). Its comparison with the compressed arithmetic word changes sign at $z = (\sqrt{11335} - 47)/39$. The figure is an auxiliary finite example, not a native period calculation. All source coordinates and phases are specified in RM22 and the reproducible builder.

For the actual growing-dimensional word, a coherent full metric enclosure $\alpha G \preceq \tilde{G} \preceq \beta G$ gives the uniform bound

$$|\log \det \mathcal{Y}_{\tilde{G}}(z) - \log \det \mathcal{Y}_G(z)| \leq (\Delta + p) \log(\beta/\alpha),$$

where $\Delta = \text{rank } T$ and $p = \text{rank}(TI_K)$. RM19–21 proves the moving-section comparison. Fixed enclosure ratios cost $O(k)$ on the original word, below its kq determinant scale.

Every original lower-root constraint in the sampling map

The row functions give physical moments $L_N[j, n] = (-i)^n n! [w^n] A_{z_j}(w)$. The matrix J_N contains all coefficient columns of multiplication by the full monic lower polynomial Q_- . The exact relation to the orthogonal projection is

$$\begin{array}{ccc} \mathcal{P}_{N-v-q'} & \xrightarrow{J_N: p \mapsto Q-p} & \mathcal{P}_{N-v} \\ & & \downarrow L_N \\ & & \mathbb{C}^r. \end{array}$$

$$C_N = (L_N J_N)(J_N^* H_N J_N)^{-1} (L_N J_N)^*, \quad B_N = L_N H_N^{-1} L_N^*.$$

Here H_N is the full physical Gamma Gram; its mass remains $\sqrt{2\pi}$. CS19 proves this map equals the full lower-root projector. CS21-23 keeps the error in every coefficient of J_N , through the original metric, before any determinant is taken. The result is a certified approximation of these complete maps. The sample count alone is not a count of bit operations, zero isolation or numerical evaluation of the native period.

Part I

The complete measured resolvent and kernel memory

The exact kernel memory of the original measured resolvent

22 September 2026. Result SZ-20260922-020, root derivation RM1–RM23.

The received PR1–PR44 calculation evaluates a determinant of the full observed resolvent. This continuation identifies the entire matrix correction produced by the observation kernel, constructs the smallest kernel space that carries that correction, and proves its exact reconstruction from the observed family. Every original metric, observation, word and complex cross term remains in the formulas. The construction applies directly to the programme's word $T_k = D_k(M)$ and its canonical metrics at the four original cutoffs. It does not assign the unresolved native determinant coefficient.

The elimination used here is the finite positive-metric Feshbach–Schur map. Its human source is Geneviève Dusson, Israel Michael Sigal and Benjamin Stamm, *The Feshbach–Schur map and perturbation theory*, original author TeX AriFestFSM_arxiv.tex, theorem thm:isospF, equations Fesh, QP, Hlam-deco, Ulam-def, source lines 398–489. The exact substitution is their operator $H - \lambda$ with our $T^{\dagger G}T + zI$, their projection with the original minimum-section projection $L\Lambda$, and $\lambda = -z$. The finite proofs below also supply the complete elimination and reconstruction explicitly. This is a use of that established construction, not a claim of inventing it.

RM1. Original value and kernel coordinates

Let $E = \mathbb{C}^q$ carry the positive Hermitian metric G , let $\Lambda : E \rightarrow B = \mathbb{C}^n$ be onto, and let $K = \ker \Lambda$, $m = q - n$. Let $I_K : \mathbb{C}^m \rightarrow E$ be a fixed full frame of the actual kernel, and retain a linear word $T : E \rightarrow E$. Put

$$Q = (\Lambda G^{-1} \Lambda^*)^{-1}, \quad L = G^{-1} \Lambda^* Q, \quad H_K = I_K^* G I_K, \quad H = T^{\dagger G} T = G^{-1} T^* G T. \quad (\text{RM1})$$

Then $\Lambda L = I_n$, $L^* G L = Q$, and $I_K^* G L = 0$. In particular the map

$$U_0 : B \oplus \mathbb{C}^m \longrightarrow E, \quad U_0(b, a) = Lb + I_K a \quad (\text{RM2})$$

is an isomorphism with source metric $\text{diag}(Q, H_K)$. Indeed applying Λ to a zero image gives $b = 0$, then injectivity of I_K gives $a = 0$; dimensions finish the proof. The orthogonality follows by direct substitution in RM1. This proves the coordinate map without replacing the value metric or kernel metric by a different source.

Define the three energy blocks

$$E_{BB} = L^* T^* G T L, \quad X = I_K^* T^* G T L, \quad E_{KK} = I_K^* T^* G T I_K. \quad (\text{RM3})$$

Thus $U_0^* T^* G T U_0 = \begin{pmatrix} E_{BB} & X^* \\ X & E_{KK} \end{pmatrix} \succeq 0$. Notice that E_{KK} is the energy of the full rectangular map $T I_K$. No compressed arithmetic endomorphism has been substituted.

RM2. The complete attained minimum at every positive parameter

For $z > 0$, set

$$D(z) = X^*(zH_K + E_{KK})^{-1}X, \quad F(z) = zQ + E_{BB} - D(z). \quad (\text{RM4})$$

The matrix $F(z)$ is positive definite. More precisely, for every $b \in B$,

$$b^*F(z)b = z b^*Qb + \min_{a \in \mathbb{C}^m} \{ \|T(Lb + I_K a)\|_G^2 + z a^*H_K a \}. \quad (\text{RM5})$$

The unique minimizer is

$$a_z(b) = -(zH_K + E_{KK})^{-1}Xb. \quad (\text{RM6})$$

To prove both statements, expand the braces using RM3, then complete the square with its strictly positive a -block. The remaining quadratic form is $E_{BB} - D(z) \succeq 0$. Adding $zQ > 0$ proves strict positivity. This also specifies the map lifting each observed vector back to its exact minimizing full vector $Lb + I_K a_z(b)$.

RM3. Matrix recovery of the full measured resolvent

Retain the actual observed family

$$\mathcal{Y}(z) = \Lambda z(zI + H)^{-1}L.$$

Solving the two block equations of $U_0^*G(zI + H)U_0$, whose right-hand side for an observed input b is $z(Qb, 0)^T$, gives

$$\boxed{\mathcal{Y}(z) = zF(z)^{-1}Q.} \quad (\text{RM7})$$

Consequently the entire memory matrix is recoverable from the original observed family:

$$\boxed{D(z) = E_{BB} - zQ(\mathcal{Y}(z)^{-1} - I).} \quad (\text{RM8})$$

The multiplication order in RM7-RM8 is essential. For example $\mathcal{Y}(z)$ need not be Hermitian in the original coefficient frame; $Q\mathcal{Y}(z)$ is Hermitian. The high-parameter coefficient also recovers E_{BB} :

$$E_{BB} = \lim_{z \rightarrow \infty} zQ(\mathcal{Y}(z)^{-1} - I), \quad (\text{RM9})$$

because $D(z) = O(z^{-1})$. All these statements are finite exact identities in the displayed maps.

RM4. The positive memory measure and its complete kernel space

For the sole purpose of expressing the same metric in an orthonormal frame, define

$$A = H_K^{-1/2}E_{KK}H_K^{-1/2}, \quad C = H_K^{-1/2}XQ^{-1/2}, \quad B = Q^{-1/2}E_{BB}Q^{-1/2}. \quad (\text{RM10})$$

The inverse coordinate maps are specified by $I_K H_K^{-1/2}$ and $LQ^{-1/2}$; thus no original mass or norm is removed. Write $A = \sum_{\lambda \geq 0} \lambda P_\lambda$, with its complete spectral projections, including multiplicities. Positivity of RM3 implies $P_0 C = 0$. One proof is that a vector $a \in \ker A$ gives zero energy to $TI_K H_K^{-1/2}a$, hence this vector is zero and its energy pairing with $TLQ^{-1/2}b$ vanishes for every b .

Therefore

$$\widehat{D}(z) := Q^{-1/2} D(z) Q^{-1/2} = \sum_{\lambda>0} \frac{M_\lambda}{z + \lambda}, \quad M_\lambda = C^* P_\lambda C \succeq 0. \quad (\text{RM11})$$

This is a finite matrix Stieltjes transform: the definition here is exactly a sum of positive matrix masses divided by $z + \lambda$ for positive real λ . Repeated kernel energies use one full residue M_λ ; no arbitrary eigenvector choice enters it. At a zero energy, the residue vanishes by the proved relation $P_0 C = 0$, rather than by deletion.

Define the actual observable kernel subspace

$$\mathcal{H}_{\text{mem}} = \text{span}\{A^j C b : b \in \mathbb{C}^n, 0 \leq j < m\} \subseteq \mathbb{C}^m. \quad (\text{RM12})$$

Cayley-Hamilton shows that this space is A -invariant. Since $A = A^*$, its orthogonal complement is invariant too. Spectral interpolation by polynomials in A then proves

$$\mathcal{H}_{\text{mem}} = \bigoplus_{\lambda>0} \text{ran}(P_\lambda C), \quad \dim \mathcal{H}_{\text{mem}} = \sum_{\lambda>0} \text{rank } M_\lambda. \quad (\text{RM13})$$

For the rank statement use $M_\lambda = (P_\lambda C)^* (P_\lambda C)$, so both matrices have the same null space. The inclusion $I_K H_K^{-1/2} : \mathcal{H}_{\text{mem}} \hookrightarrow K \subset E$ is the exact connecting isometry to the programme. The orthogonal complement is retained as a reducing kernel space with zero coupling to the observation.

RM5. Minimality and uniqueness, with a finite moment certificate

Any representation of RM11 in the form $C_1^* (zI + A_1)^{-1} C_1$, with $A_1 \succeq 0$ and no zero residue, has a residue $C_1^* P_{1,\lambda} C_1 = M_\lambda$ at each actual pole. The dimension of its λ -eigenspace is at least $\text{rank } M_\lambda$. Summing gives the lower bound in RM13. Restriction to $\mathcal{H}_{\text{mem}}$ attains it, so this is a smallest realization of the original memory.

It is unique up to an isometry that also intertwines A and C . Here is a direct proof. Equality of the rational functions gives equality of their expansions at infinity, hence

$$C^* A^j C = C_1^* A_1^j C_1 \quad (j \geq 0).$$

Send each vector $\sum_j A^j C b_j$ to $\sum_j A_1^j C_1 b_j$. The squared norm and every cross inner product are the same, by the moment identities with exponent $i + j$. The map is well-defined, isometric, onto the generated spaces, and intertwines A and A_1 . It also maps C to C_1 . This proves the assertion.

There is an exact finite rank certificate. The block matrix

$$\mathcal{M}_m = [C^* A^{i+j} C]_{i,j=0}^{m-1}$$

is the Gram matrix of $[C, AC, \dots, A^{m-1}C]$. Thus

$$\boxed{\text{rank } \mathcal{M}_m = \dim \mathcal{H}_{\text{mem}}.} \quad (\text{RM14})$$

It uses the original phase-bearing moments, not their scalar traces.

RM6. The exact correction to compressing before taking the resolvent

In the orthonormal value frame the compression of the positive energy H is B . Put

$$F_0 = zI + B, \quad R(z) = F_0^{-1/2} \widehat{D}(z) F_0^{-1/2}.$$

RM5 gives $0 \preceq R(z) \prec I$. Comparing the correct family with $Y_{\text{energy}}(z) = z(zI + B)^{-1}$ gives

$$\widehat{\mathcal{Y}}(z) \succeq Y_{\text{energy}}(z), \quad \log \frac{\det \mathcal{Y}(z)}{\det Y_{\text{energy}}(z)} = -\log \det(I - R(z)) \geq 0. \quad (\text{RM15})$$

Both statements follow by inversion and the factorization of $F_0 - \widehat{D}$. The rank of $R(z)$ equals $\text{rank } C$, and the exact correction is the sum $-\sum_{j=1}^{\text{rank } C} \log(1 - r_j(z))$. In particular it vanishes at one positive z if and only if $C = 0$, which is equivalent to K reducing the positive energy H . This proves precisely when the omission is allowed.

For comparison with the arithmetic operator compressed first, write the full word itself in the same orthogonal splitting:

$$\tilde{T} = \begin{pmatrix} T_{BB} & T_{BK} \\ T_{KB} & T_{KK} \end{pmatrix}.$$

Then

$$B = T_{BB}^* T_{BB} + T_{KB}^* T_{KB}, \quad C = T_{BK}^* T_{BB} + T_{KK}^* T_{KB}, \quad A = T_{BK}^* T_{BK} + T_{KK}^* T_{KK}. \quad (\text{RM16})$$

Substitution in RM7 proves the complete relation to the resolvent of $T_{BB}^* T_{BB}$: its missing effective term is $T_{KB}^* T_{KB} - C^*(zI + A)^{-1}C$. Its sign is not fixed. Indeed with scalar blocks, $T = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ gives positive term 1, while $T = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ gives negative term $-1/(z+1)$. These exact words have positive full energies. Thus the defect defines the specified memory family; it is not a reason to end the calculation.

RM7. Every zero direction and every determinant factor

The full energy in this frame is $\begin{pmatrix} B & C^* \\ C & A \end{pmatrix}$. Let $\mathcal{H}_{\text{inv}} = \mathcal{H}_{\text{mem}}^\perp \subset K$. It is reducing and has no coupling to B . Hence

$$\det(zI + H) = \det(zI + A|_{\mathcal{H}_{\text{inv}}}) \det \begin{pmatrix} zI + B & C_{\text{mem}}^* \\ C_{\text{mem}} & zI + A_{\text{mem}} \end{pmatrix}.$$

The same factor occurs in $\det(zI + A)$ in the numerator of PR7 and cancels exactly. It includes every zero direction of $K \cap \ker T$, since those are in $\mathcal{H}_{\text{inv}}$. The invariant space and its energy are retained explicitly; they have zero residue in RM11. The cancellation therefore has a proved map and a precise mathematical reason.

The remaining determinant identity is

$$\det \mathcal{Y}(z) = z^n \frac{\det(zI + A_{\text{mem}})}{\det \begin{pmatrix} zI + B & C_{\text{mem}}^* \\ C_{\text{mem}} & zI + A_{\text{mem}} \end{pmatrix}}. \quad (\text{RM17})$$

It is the PR determinant formula with all unobserved reducing factors canceled by exact equality, not by an asymptotic allocation.

RM8. Application to the original mixed arithmetic family

For each original $N \in \{q-1, q, 2q-1, 2q\}$, use $T = T_k = D_k(M)$, the unchanged G_N, Λ, I_K , and $m = 8k - 16$. Then RM4 is the explicit original-metric matrix

$$D_N(z) = L_N^* T_k^* G_N T_k I_K (z H_{K,N} + H_{TK,N})^{-1} I_K^* T_k^* G_N T_k L_N. \quad (\text{RM18})$$

The received PR19 gives $H_{TK,N} = J^* \mathfrak{G}_N J$ with the fixed exact map $J = V_O^{-1} \pi_{\partial} I_K$; this is also the image map proved in the full-word calculation. Substituting that image Gram in RM18 leaves the complete original $H_{K,N}$ and both complex cross blocks. The positive word spectrum alone does not determine RM18, whereas the full measured family determines it by RM8-RM9.

The finite sign and deep-regularizer results PR25-PR31 are preserved at their stated analytic input status. RM18 neither replaces their native kernel coefficient nor derives a sign of a different complex current. It supplies the exact matrix object that carries the mixed kernel/observation coupling, and RM12-RM14 give a concrete finite construction and rank test for all of its contributing directions. These are the next matrices to evaluate with the complete-row sampling bounds; a determinant bound on them does not erase the off-diagonal terms.

RM9. Coherent error in the original full source metric

Keep exactly the same linear maps T, Λ, I_K . Let G and $\tilde{G}$ be two positive Hermitian forms on that same original space, and let positive numbers $\alpha \leq \beta$ satisfy

$$\alpha G \preceq \tilde{G} \preceq \beta G, \quad \kappa = \beta/\alpha.$$

Every object bearing a tilde is recomputed from that full form by RM1-7; in particular its minimum section is recomputed, rather than held fixed when its defining metric changes. Write $\Delta = \text{rank } T$, $p = \text{rank}(T I_K)$, and let $\lambda_1, \dots, \lambda_{\Delta} > 0$ be the positive eigenvalues of $G^{-1} T^* G T$, in decreasing order. Let $a_1, \dots, a_p > 0$ be the positive generalized eigenvalues of $(I_K^* T^* G T I_K, I_K^* G I_K)$. The remaining eigenvalues in both lists are exactly zero and their multiplicities depend only on the fixed maps.

For every vector $x \neq 0$, the two Rayleigh quotients of the full positive energy satisfy

$$\kappa^{-1} \frac{\|Tx\|_G^2}{\|x\|_G^2} \leq \frac{\|Tx\|_{\tilde{G}}^2}{\|x\|_{\tilde{G}}^2} \leq \kappa \frac{\|Tx\|_G^2}{\|x\|_G^2}.$$

The identical proof for $x = I_K b$ applies on the complete kernel. The min-max formula now gives, in the same ordering,

$$\kappa^{-1} \lambda_j \leq \tilde{\lambda}_j \leq \kappa \lambda_j, \quad \kappa^{-1} a_j \leq \tilde{a}_j \leq \kappa a_j. \quad (\text{RM19})$$

For clarity, this use of min-max is finite: choose an orthonormal eigenbasis for either Hermitian representative, express a vector's quotient as its eigenvalue-weighted squared coordinates, and maximize the minimum over subspaces of prescribed dimension. Intersection with the span of the smaller eigenvectors proves the upper bound, while the span of the larger eigenvectors attains equality. This argument permits repeated eigenvalues. Applying the pointwise quotient comparison on every such subspace proves RM19.

The original block determinant in RM7, before any reducing-factor cancellation, is

$$\det \mathcal{Y}_G(z) = z^{\Delta-p} \frac{\prod_{j=1}^p (z + a_j)}{\prod_{j=1}^{\Delta} (z + \lambda_j)}.$$

Indeed its numerator is $z^n \det(zI_m + A)$, its denominator is $\det(zI_q + H)$, and $n = q - m$; the zero multiplicities therefore give $z^{n+m-p-(q-\Delta)} = z^{\Delta-p}$. The identical power occurs for $\tilde{G}$. By RM19, each remaining positive factor changes by a ratio in $[\kappa^{-1}, \kappa]$ for every $z > 0$. Thus

$$\boxed{|\log \det \mathcal{Y}_{\tilde{G}}(z) - \log \det \mathcal{Y}_G(z)| \leq (\Delta + p) \log \kappa.} \quad (\text{RM20})$$

This bound is uniform in the regularizer, including values smaller than every positive energy. It counts the actual nonzero energies and keeps all exact zero factors before canceling them. It also includes the metric-induced change of the original section, since both determinants were derived with their own RM1 section.

For the programme word, $\Delta = 16k - 48$, $p = m = 8k - 16$, by its proved trivial intersection $K \cap \ker T_k = 0$. Apply RM20 separately to its four original full forms, using their own positive enclosures α_N, β_N , at any chosen positive parameters z_N . The exact four-sign difference is bounded by

$$\boxed{\left| \mathcal{R} \log \det \mathcal{Y}_{\tilde{G}_N}(z_N) - \mathcal{R} \log \det \mathcal{Y}_{G_N}(z_N) \right| \leq (24k - 64) \sum_{N \in \{q-1, q, 2q-1, 2q\}} \log(\beta_N / \alpha_N).} \quad (\text{RM21})$$

The sum is over four cutoffs, with the return signs $+, +, -, -$ unchanged in the left side. A fixed enclosure ratio costs $O(k) = o(kq)$. More generally the bound is $o(kq)$ whenever the displayed sum divided by q tends to zero. This is an error theorem for the complete form and its actual maps; the CS20-23 enclosure of a particular row covariance must still be transported by its exact defining maps before it can serve as a full-form enclosure. In both constructions the common operation is congruence followed by a full minimum and inverse. RM19-21 prove this operation for the measured resolvent itself, including its moving section. No native coefficient is inferred from an enclosure width.

RM10. An exact illustration of the phase-bearing correction

Take the finite auxiliary example

$$G = I_3, \quad T = \text{diag}(0, 2, 3i), \quad \Lambda = \begin{pmatrix} -1 & 1 & 0 \\ -i & 0 & 1 \end{pmatrix}, \quad I_K = (1, 1, i)^\top.$$

Direct substitution in RM1-4 gives

$$Q = \frac{1}{3} \begin{pmatrix} 2 & i \\ -i & 2 \end{pmatrix}, \quad L = \frac{1}{3} \begin{pmatrix} -1 & i \\ 2 & i \\ -i & 2 \end{pmatrix}, \quad H_K = 3, \quad E_{KK} = 13, \quad X = \frac{1}{3}(-1, -14i),$$

$$E_{BB} = \frac{1}{9} \begin{pmatrix} 25 & 26i \\ -26i & 40 \end{pmatrix}, \quad D(z) = \frac{1}{9(3z + 13)} \begin{pmatrix} 1 & 14i \\ -14i & 196 \end{pmatrix}.$$

Thus the memory has one positive pole energy $13/3$, its contributing kernel dimension is exactly one, and its complex off-diagonal entry is nonzero. The actual measured determinant and the two possible early compressions are

$$\det \mathcal{Y}(z) = \frac{z(3z + 13)}{3(z + 4)(z + 9)},$$

$$\det Y_{\text{energy}}(z) = \frac{3z^2}{3z^2 + 26z + 36}, \quad \det Y_{\text{arithmetic}}(z) = \frac{9z^2}{9z^2 + 52z + 36}.$$

Here the second uses the positive energy compression $Q^{-1}E_{BB}$, while the third uses the square of the actual compressed word $T_{BB} = \Lambda TL$ in its original metric Q . The first difference is nonnegative by RM15. For the last comparison, exact subtraction gives

$$\det \mathcal{Y}(z) - \det Y_{\text{arithmetic}}(z) = \frac{-2z(39z^2 + 94z - 234)}{3(z+4)(z+9)(9z^2 + 52z + 36)}. \quad (\text{RM22})$$

All denominator factors are positive for $z > 0$. The difference is positive below $(\sqrt{11335} - 47)/39$, zero there, and negative above it. Thus its sign change is calculated on the same original maps. RM4 and RM16 provide the exact connecting correction in both cases. This example illustrates those identities; it is not a sampled native period or a claim about its current sign.

RM11. A complete complex-response error in the original value frame

The same coherent source comparison also controls every complex matrix entry of the measured action. Take $(1 - \varepsilon)G \preceq \tilde{G} \preceq (1 + \varepsilon)G$, $0 \leq \varepsilon < 1$, with the same original maps. Define the two complete compressed inverse forms

$$\Sigma_z = \Lambda(zG + T^*GT)^{-1}\Lambda^*, \quad \Sigma_0 = \Lambda G^{-1}\Lambda^* = Q^{-1},$$

and define their tilded versions by the same expressions. The exact original-frame identity is $\mathcal{Y} = z\Sigma_z\Sigma_0^{-1}$. It follows directly from $(zI + G^{-1}T^*GT)^{-1}G^{-1} = (zG + T^*GT)^{-1}$, so this expression retains the changed minimum section automatically.

Congruence by T , addition of zG , inverse order and congruence by Λ prove

$$\frac{\Sigma_j}{1 + \varepsilon} \preceq \tilde{\Sigma}_j \preceq \frac{\Sigma_j}{1 - \varepsilon}, \quad j \in \{0, z\}.$$

Put $Y = Q^{1/2}\mathcal{Y}Q^{-1/2}$, $A = zQ^{1/2}\tilde{\Sigma}_zQ^{1/2}$, and $S = Q^{1/2}\tilde{\Sigma}_0Q^{1/2}$. Then $0 \prec Y \preceq I$, $\|A - Y\| \leq \varepsilon\|Y\|/(1 - \varepsilon)$, $\|S^{-1}\| \leq 1 + \varepsilon$, and $\|S^{-1} - I\| \leq \varepsilon$. These follow from the displayed form inequalities and their inverse; no separately chosen square-root coordinate frame is compared. The actual tilded action expressed in this same original value frame is $Q^{1/2}\tilde{\mathcal{Y}}Q^{-1/2} = AS^{-1}$. Therefore

$$\left\| Q^{1/2}(\tilde{\mathcal{Y}}(z) - \mathcal{Y}(z))Q^{-1/2} \right\| \leq \frac{2\varepsilon}{1 - \varepsilon} \left\| Q^{1/2}\mathcal{Y}(z)Q^{-1/2} \right\| \leq \frac{2\varepsilon}{1 - \varepsilon}. \quad (\text{RM23})$$

Indeed subtract Y from AS^{-1} as $(A - Y)S^{-1} + Y(S^{-1} - I)$, take operator norms and add the two exact bounds. For any original observed vectors u, v , Cauchy-Schwarz now gives

$$|u^*Q(\tilde{\mathcal{Y}} - \mathcal{Y})v| \leq \frac{2\varepsilon}{1 - \varepsilon} \|Y\| \|u\|_Q \|v\|_Q.$$

This is an absolute complex-response error, with real and imaginary parts both retained. It is uniform over all positive regularizers. It does not determine the sign of a response whose actual value has not been evaluated; it gives the exact original-metric error to be paid when that value is computed. RM20 separately gives the stronger nonzero-rank determinant bound, so the phase estimate is not used to charge a full ambient rank to the determinant.

Proof and source coverage

The received COMPLETE_PROOFS (5).md, SHA-256 1e042cf77ef8df20bcb3167d38d5a98e9f653b6a2a1694a16e6c2728b7de96ab, PR1-44 was read completely. The matching resolvent paste was read completely. The derivation above proves RM1-23 in full finite dimension independently of the incoming numerical receipt. Its checker tests the complete original coordinates, nonidentity complex metrics, nonnormal words, kernel intersections, both signs of the arithmetic-compression defect, and the minimal memory moment rank. Native asymptotic coefficients are not outputs of these auxiliary checks.

The exact programme specialization is FW1-10 and FW39-49, retaining its full original root polynomial, source metric and complete observation. The finite source receiver is FI1-17, FR1-4. Their entire proofs accompany this edition.

Part II

Relative growth and growing pole removal

Relative source growth and expanding conductor-pole cancellation

Independent derivation, 22 September 2026. This continuation proves the RH-labelled sections 1–3 of the two supplied arrivals in the original invariant-row problem. The label identifies that programme; no conclusion that the Riemann hypothesis is closed is made. The residual coefficient at order kq remains a determinant to evaluate. RG10 repairs a radius guard that cannot be inferred for every allowed L_k .

The complete predecessor proofs are retained in this source collection and published: AS1-64, original angle spectrum, RX1-20, full-root innovations, and FI1-13, finite invariant receiver. The local complete-basis sampling proof CS1-24 is retained at `relative_growth_arrival_20260922/sampling_derivation/SAMPLING_PROOFS.md`. Precise versions, actual reading coverage and original human sources are recorded in `SOURCE_USE_LEDGER.md`. All new relative-growth, complementary-minimum, chronological, radius, spectral-law and common-circle sampling arguments are proved below.

RG1. Original objects, coordinates and finite domain

Fix the original admitted simple quartet, nonzero conductor and period. Put

$$\begin{aligned} k &\equiv 1 \pmod{4}, \quad k \geq 17, \quad q = (k+1)^2, \quad k' = k-8, \\ q' &= (k-7)^2, \quad 0 < \delta < 1/2, \quad \gamma > 2, \\ R_h &= \sqrt{\delta^2 + \gamma^2}, \quad q \geq 2kR_h. \end{aligned} \tag{RG1}$$

The original exponents and physical coordinates are

$$\begin{aligned} c &= k/2, \quad c' = k/2 - 4, \quad \beta_{ab} = c + (2a - k)\delta + i(2b - k)\gamma, \\ \beta'_{ab} &= c' + (2a - k')\delta + i(2b - k')\gamma, \quad \omega'_{ab} = (\beta'_{ab} - c')/i = (2b - k')\gamma - i(2a - k')\delta. \end{aligned}$$

Retain the complete original symbol

$$\begin{aligned} E_A(w) &= \sum_{a,b=0}^8 a_{ab} e^{b_{ab}w}, \quad b_{ab} = 4 + (2a - 8)\delta + i(2b - 8)\gamma, \\ v &= \text{ord}_0 E_A \leq 80, \quad \mu_v = E_A^{(v)}(0) \neq 0, \quad d_A = |\mu_v|/v!, \quad A_1 = \sum |a_{ab}|, \quad B_A = \max_{a_{ab} \neq 0} |b_{ab}|. \end{aligned}$$

The bound on v follows from the invertible Vandermonde matrix of at most 81 distinct exponents. The complete original invariant rows, after all low-polynomial constraints, form V_k , supported on the original conductor strip of size $n_E = q - q' = 16k - 48$. Let Z be its fixed full row frame,

$$R_Z = ZZ^* > 0, \quad r = 8k - 32 - v + s_k, \quad s_k = \dim(K_k \cap \mathcal{P}_{<v}), \quad 0 \leq s_k \leq v.$$

In particular $24 \leq r \leq 8k - 32$. A physical row $z \in V_k$ acts through

$$F_z(w) = \sum_{\alpha} z_{\alpha} e^{\beta_{\alpha} w}, \quad A_z(w) = e^{-c'w} F_z(w) / E_A(w), \quad \ell_z[y^n] = (-i)^n n! [w^n] A_z(w). \tag{RG2}$$

The numerator's first v derivatives vanish, so A_z is removable at zero. The identity in (RG2) is exactly $\ell_z[f] = [f((\partial_w - c')/i)(F_z/E_A)]_{w=0}$, by differentiating $e^{-c'w}$. Neither the center nor the phase is changed.

For every integer $q-1 \leq N \leq 2q$, set $D = N - v$, $M = D - q'$. Retain

$$Q_-(y) = i^{-q'} \chi'(c' + iy) = \prod_{a,b=0}^{k'} (y - \omega'_{ab}), \quad I_N = Q_- \mathbb{C}[y]_{\leq M},$$

$$d\sigma(y) = |\Gamma(1/4 + iy/2)|^2 dy / (2\pi), \quad M_\sigma = \sqrt{2\pi},$$

$$p_{n+1} = yp_n - n(n-1/2)p_{n-1}, \quad h_n = M_\sigma n!(1/2)_n.$$

with $p_0 = 1, p_1 = y$. The matrix E_D has every evaluation row $p_n(\omega'_\alpha)/\sqrt{h_n}$. Its rank is q' by interpolation; its kernel is exactly I_N , since all roots are distinct. Thus

$$P_N = I - E_D^*(E_D E_D^*)^{-1} E_D, \quad C_N = W_N P_N W_N^*, \quad B_N = W_N W_N^*, \quad S_N = B_N^{-1/2} C_N B_N^{-1/2}. \quad (\text{RG3})$$

Here W_N comprises the original functionals on $p_n/\sqrt{h_n}$. The fixed coefficient congruence is $C_N^\circ = R_Z^{-1/2} C_N R_Z^{-1/2}$, $B_N^\circ = R_Z^{-1/2} B_N R_Z^{-1/2}$. It compares the same rows and leaves the physical source unchanged. Its unitary relation with the physical angle matrix is induced by $B_N^{1/2} R_Z^{-1/2} (B_N^\circ)^{-1/2}$. In the subsequent matrix arguments C_N, B_N abbreviate these congruent matrices. The inverse congruence recovers the physical frame.

RG2. Explicit previously proved finite constants

These are the finite constants of AS6-31, with no unspecified remainder in a finite inequality. Let $a_* \neq 0$ be the actual lexicographic conductor pivot and $C_{\text{path}} = A_1/|a_*|$. Put

$$\mathfrak{M} = \max\{1, e^{1+2\pi\gamma} A_1 (8q+1)^{1/4} (2q+1)(4q+1)^{4\delta}\},$$

$$B_{\text{val}} = q M_\sigma e^{8q} \left(\frac{2q + kR_h}{2\delta k} \right)^{2(q-1)}, \quad B_E = n_E M_\sigma \left(\frac{2n_E + kR_h}{2\delta} \right)^{2(n_E-1)},$$

$$c_k = (\mathfrak{M}^2 B_{\text{val}} q C_{\text{path}}^{20k})^{-1}, \quad b_k^- = (\mathfrak{M}^2 B_E)^{-1}. \quad (\text{RG4})$$

AS6-11 prove $C_N \succeq c_k I$ using the full upper-root product, the actual strip elimination and the complete lower-root minimum. AS26 proves $B_N \succeq b_k^- I$ using the entire strip. The exact expansion is

$$\log(1/c_k) = 2(q-1) \log(q/k) + 8q + 2(q-1) \log \frac{2 + kR_h/q}{2\delta} + 2 \log \mathfrak{M} + \log(q^2 M_\sigma) + 20k \log C_{\text{path}}. \quad (\text{RG5})$$

No monic-profile asymptotic is used.

Define

$$a_0 = \frac{\log^+(A_1/d_A)}{\log 2}, \quad b_0 = \frac{4B_A}{\log 2}, \quad C_A = \frac{\max(A_1, d_A)^2}{d_A^3}, \quad M_k(R) = C_A \sqrt{n_E} e^{(2kR_h + 8 + 4B_A)R},$$

$$\bar{B}_A = \max(1, B_A), \quad \rho_A = \min \left\{ 1, \frac{(v+1)! d_A}{2A_1 \bar{B}_A^{v+1} e^{\bar{B}_A}} \right\}, \quad M_{0,k} = 2\sqrt{n_E} \rho_A^{-v} d_A^{-1} e^{(4+kR_h)\rho_A}.$$

Put $M_{\text{max}} = 2q - v - q'$, $c_\sigma = \Gamma(1/4)^2/(2\pi)$, and

$$\mathfrak{A} = (M_{\text{max}} + 1) 10^{M_{\text{max}}} 2^{q'} e^{\pi q/2} \sqrt{\frac{1 + 16q^2}{c_\sigma q}},$$

$$\mathfrak{F} = \mathfrak{A}C_A\sqrt{n_E}e^{2R_hq+(8+4B_A)k}(R_h+2)^{q'}, \quad \mathfrak{F}_0 = \mathfrak{A}M_{0,k}(kR_h/q+2/\rho_A)^{q'}(2/\rho_A)^{M_{\max}},$$

$$L_A = b_0 + 9, \quad j_A = \lceil \max\{2(a_0 + 1), a_0 + 1 + 4L_A\} \rceil, \quad \mathfrak{U}_k = \max\{\mathfrak{F}^2(4L_A)^{2q'}, \mathfrak{F}_0^2 j_A^{2q'}\}. \quad (\text{RG6})$$

AS12–25 prove, at every cutoff,

$$\lambda_j^\perp(C_N) \leq \mathfrak{U}_k j^{-2q'}, \quad 1 \leq j \leq r. \quad (\text{RG7})$$

The exact index is justified by $R_j = (j - 1 - a_0)/(2L_A)$: for $j \geq j_A$, $2 \leq R_j \leq k$, $R_j \geq j/(4L_A)$, and the true pole-constraint codimension is $< j$. The first j eigenspace therefore intersects that subspace. For $j < j_A$, the fixed-radius bound supplies the second term of (RG6). This also covers $j_A > r$.

For the angle constant set $t_A = \tanh(\rho_A/2)$,

$$\begin{aligned} b_k^+ &= \max\{1, M_{0,k}^2 M_\sigma^{-1} (1 - t_A^2)^{-1/2} (2q + 1)^2 t_A^{-4q}\}, \\ \ell_k &= \log(q/k), \\ C_{\text{ang}} &= \max \left\{ \begin{array}{l} 0, -q^{-1} \log(c_k/b_k^+) - 2\ell_k, \\ q^{-1} \log(\mathfrak{U}_k/b_k^-) + 2\ell_k - \frac{2q'}{q} \log r \end{array} \right\}. \end{aligned} \quad (\text{RG8})$$

AS29–31 give

$$-C_{\text{ang}} \leq x_{j,N} := q^{-1} \log s_{j,N} + 2\ell_k \leq C_{\text{ang}} + 2 \log(r/j). \quad (\text{RG9})$$

For fixed data, (RG5) gives $\log(1/c_k) = 2(q - 1)\ell_k + O(q)$, (RG6) gives $\log \mathfrak{U}_k = O(q)$, and $\log(1/b_k^-) = O(k \log q)$. Also $\log b_k^+ = O(q)$, and $2\ell_k - 2(q'/q) \log r = O(1)$, since $r = 8k + O(1)$. Hence $C_{\text{ang}} = O(1)$. None of these statements is asserted uniformly through degeneration of d_A or a_* .

RG3. Exact full-root chronological innovation

Append one original orthonormal source column: $E_{D+1} = [E_D, e]$, $W_{N+1} = [W_N, w]$. Let $G = E_D E_D^*$. In the new source,

$$t = \frac{1}{\sqrt{\kappa}} \begin{pmatrix} -E_D^* G^{-1} e \\ 1 \end{pmatrix}, \quad \kappa = 1 + e^* G^{-1} e$$

has norm one, lies in the full evaluation kernel, and is perpendicular to the embedded old kernel. The new kernel has dimension exactly one more. Consequently

$$C_{N+1} = C_N + \eta_N \eta_N^*, \quad \eta_N = \frac{w - W_N E_D^* G^{-1} e}{\sqrt{\kappa}}, \quad B_{N+1} = B_N + w w^*. \quad (\text{RG10})$$

This reproves RX1 with every complex cross term and every root retained. Congruence multiplies both new columns by $R_Z^{-1/2}$.

RG4. Relative-growth spectrum, tails and all exterior ranks

Take any integers $q - 1 \leq l \leq u \leq 2q$, and define

$$\Gamma = C_l^{-1/2} C_u C_l^{-1/2}, \quad \mu_1 \geq \cdots \geq \mu_r \geq 1, \quad \Omega = \log(\mathfrak{U}_k/c_k), \quad \alpha = q'/q, \quad B_{\text{inc}} = \frac{\Omega - 2q' \log r}{q}. \quad (\text{RG11})$$

The eigenvalues of Γ equal those of $C_u^{1/2}C_l^{-1}C_u^{1/2}$, because TT^* and T^*T have identical spectra for the invertible matrix $T = C_l^{-1/2}C_u^{1/2}$. The latter matrix is at most $c_k^{-1}C_u$. Min-max and (RG7) therefore give

$$1 \leq \mu_j \leq (\mathfrak{U}_k/c_k)j^{-2q'}, \quad 0 \leq q^{-1} \log \mu_j \leq B_{\text{inc}} + 2\alpha \log(r/j). \quad (\text{RG12})$$

The inequality $c_k \leq \lambda_r(C_l) \leq \mathfrak{U}_k r^{-2q'}$ proves $B_{\text{inc}} \geq 0$ exactly. From (RG5), $qB_{\text{inc}} = 2(q-1) \log(q/k) - 2q' \log r + O(q) = O(q)$; hence $B_{\text{inc}} = O(1)$.

For $T \geq B_{\text{inc}}$, count the indices in (RG12) and integrate:

$$\frac{1}{r} \sum_j (q^{-1} \log \mu_j - T)_+ \leq 2\alpha e^{-(T-B_{\text{inc}})/(2\alpha)}. \quad (\text{RG13})$$

For $0 \leq \theta < (2\alpha)^{-1}$, comparison of $\sum j^{-2\alpha\theta}$ with its integral on $(0, r]$ gives $r^{-1} \sum_j \mu_j^{\theta/q} \leq e^{\theta B_{\text{inc}}}/(1 - 2\alpha\theta)$.

For every integer $0 \leq h \leq r$, define

$$\mathfrak{D}_k(0) = 0, \quad \mathfrak{D}_k(h) = h\Omega - 2q' \log(h!) = qhB_{\text{inc}} + 2q' \log(r^h/h!). \quad (\text{RG14})$$

Then

$$\log \|\wedge^h \Gamma\| = \sum_{j=1}^h \log \mu_j \leq \mathfrak{D}_k(h) \leq qh[B_{\text{inc}} + 2\alpha(1 + \log(r/h))]. \quad (\text{RG15})$$

The last inequality is $\log(h!) \geq h \log h - h$, obtained by integrating $\log t$. Also $\mathfrak{D}_k(h) - \mathfrak{D}_k(h-1) = \Omega - 2q' \log h \geq 0$. Since $t \log(1/t) \rightarrow 0$, $h = o(k)$ implies $\mathfrak{D}_k(h) = o(kq)$. The bounds are uniform over the complete integer cutoff window.

RG5. Restrictions and attained quotients: complete generalized interlacing

Let $A \preceq D$ be positive definite forms on $\mathbb{C}^r$, with relative eigenvalues μ_j in decreasing order. For a fixed injection $J : \mathbb{C}^m \rightarrow \mathbb{C}^r$, $d = r - m$, the matrix

$$V = A^{1/2}J(J^*AJ)^{-1/2}$$

is an isometry, and the relative restriction matrix equals

$$(J^*AJ)^{-1/2}J^*DJ(J^*AJ)^{-1/2} = V^*(A^{-1/2}DA^{-1/2})V.$$

Its ordered eigenvalues satisfy

$$\mu_j \geq \theta_j^J \geq \mu_{j+d}, \quad 1 \leq j \leq m. \quad (\text{RG16})$$

The upper inequality follows by restricting the maximizing j -dimensional trial spaces to $\text{im } V$. For the lower inequality, the top $(j+d)$ -dimensional eigenspace intersects this codimension- d image in dimension at least j , where the quadratic form is at least μ_{j+d} . This proves the finite min-max statement without eigenvector alignment.

For any fixed surjection $\pi : \mathbb{C}^r \rightarrow \mathbb{C}^m$, define

$$Q_A = (\pi A^{-1} \pi^*)^{-1}, \quad \min_{\pi x=y} x^* A x = y^* Q_A y, \quad x_{\min} = A^{-1} \pi^* Q_A y. \quad (\text{RG17})$$

The proposed point maps to y ; every other point differs by $z \in \ker \pi$, and its cross term vanishes since $Ax_{\min} = \pi^* Q_A y$. Positivity proves existence and uniqueness of

the attained minimum. Increasing the form increases its minimum on each fixed fibre; thus $Q_A \preceq Q_D$.

The matrix $V = A^{-1/2} \pi^* Q_A^{1/2}$ is an isometry and

$$V^*(A^{-1/2} D A^{-1/2})^{-1} V = Q_A^{1/2} Q_D^{-1} Q_A^{1/2} = (Q_A^{-1/2} Q_D Q_A^{-1/2})^{-1}.$$

Apply compression interlacing to the inverses and invert their ordered eigenvalues. The attained-quotient relative eigenvalues satisfy

$$\mu_j \geq \theta_j^Q \geq \mu_{j+d}, \quad 1 \leq j \leq m. \quad (\text{RG18})$$

This inversion step specifies the exact quotient map; compressing A itself would instead produce its restriction.

RG6. Complete complementary Schur loss

Choose one fixed invertible coefficient frame $[J, K]$, with J the retained rank- $r - h$ inclusion. In this frame write

$$C_N = \begin{pmatrix} A_N & T_N \\ T_N^* & D_N \end{pmatrix}, \quad \Sigma_N = D_N - T_N^* A_N^{-1} T_N.$$

Completion of the square proves

$$\det C_N = \det A_N \det \Sigma_N,$$

$$u^* C_N^{-1} v = u_1^* A_N^{-1} v_1 + (u_2 - T_N^* A_N^{-1} u_1)^* \Sigma_N^{-1} (v_2 - T_N^* A_N^{-1} v_1). \quad (\text{RG19})$$

For a nonunitary frame its constant squared determinant is present at each cutoff and cancels between the same endpoints. The form Σ_N is exactly the attained quotient under projection onto the complementary h entries, and therefore increases with N .

Write $G(l, u) = \log(\det C_u / \det C_l)$, $G_J(l, u) = \log(\det A_u / \det A_l)$. Equations (RG18)–(RG19) prove

$$0 \leq G(l, u) - G_J(l, u) = \log \frac{\det \Sigma_u}{\det \Sigma_l} \leq \sum_{j=1}^h \log \mu_j \leq \mathfrak{D}_k(h). \quad (\text{RG20})$$

For a retained attained quotient of rank $r - h$, the missing determinant is the restriction to its rank- h kernel. The same square completion and (RG16) prove the identical bound. Every complementary minimum is retained, and no inverse cross term in (RG19) has been set to zero.

RG7. Selected actual chronological increments

Let $\mathcal{J} \subset \{l, \dots, u - 1\}$ contain h selected original additions. Put $d_N = \log(\det C_{N+1} / \det C_N) = \log(1 + \eta_N^* C_N^{-1} \eta_N) \geq 0$. The selected-only partial sums $C_N^{\mathcal{J}} = C_l + \sum_{j < N, j \in \mathcal{J}} \eta_j \eta_j^*$ satisfy $C_N^{\mathcal{J}} \preceq C_N$, hence $(C_N^{\mathcal{J}})^{-1} \succeq C_N^{-1}$. The determinant lemma at every selected step gives

$$\sum_{N \in \mathcal{J}} d_N \leq \log \frac{\det(C_l + \sum_{N \in \mathcal{J}} \eta_N \eta_N^*)}{\det C_l}.$$

After whitening, the latter relative matrix lies between I and Γ and differs from I by rank at most $\min(h, r)$. Min-max bounds its nonunit eigenvalues by the first $\min(h, r)$ eigenvalues of Γ . Therefore

$$0 \leq \sum_{N \in \mathcal{J}} d_N \leq \mathfrak{D}_k(\min(h, r)). \quad (\text{RG21})$$

These are the actual increments in the full chronology. Their sum generally differs from the selected-only determinant gain; the proved inequality is the needed connecting map. On either original q -step window, $o(k)$ selected additions contribute $o(kq)$.

For any one fixed retained rank- s row inclusion, the updates are exactly $J^* \eta_N \eta_N^* J$. The same proof on that row space, followed by (RG16), gives the sharper rank cap $\mathfrak{D}_k(\min(h, s))$ for its selected actual increments. This statement uses the retained chronological covariance at each step, including all earlier additions to that row space.

RG8. Actual pole jets and the finite holomorphic envelope

Set $b(w) = E_A(w)/w^v$, so b is entire and $|b(0)| = d_A$. Jensen's formula on radius $4R$, obtained by factoring its finitely many interior zeros and applying the mean-value identity to the remaining logarithm, gives for $R \geq 1$

$$n_b(2R) \log 2 \leq \log^+(A_1/d_A) + 4B_A R. \quad (\text{RG22})$$

The boundary estimate is $|b(w)| \leq A_1(4R)^{-v} e^{4B_A R} \leq A_1 e^{4B_A R}$; approach boundary-zero radii by admissible radii. Zeros in the open disk are counted with complete multiplicities.

Let $\mathcal{Z}_R \subset V_k$ be the kernel of all original equations $F_z^{(j)}(\zeta) = 0$ for these zeros and $0 \leq j < \text{ord}_\zeta b$. Its actual codimension obeys

$$h_R \leq n_b(2R) \leq a_0 + b_0 R. \quad (\text{RG23})$$

These kernels are nested in R , fixed across N , and no independence of their equations is assumed.

Form the radius- $2R$ finite Blaschke product

$$\mathcal{B}_R(w) = \prod_{\zeta} \left(\frac{2R(w - \zeta)}{(2R)^2 - \bar{\zeta}w} \right)^{m_\zeta}.$$

It has modulus one on the boundary and its poles are outside the closed disk. Both $b/\mathcal{B}_R$ and $e^{-c'w} F_z/(w^v \mathcal{B}_R)$ extend across every listed zero; the first has no zero in the open disk. Put $M_b = \max(A_1, d_A) e^{2B_A R}$. The maximum principle gives $|b/\mathcal{B}_R| \leq M_b$, and its value at zero has modulus at least d_A . The nonnegative harmonic function $u = \log(M_b/|b/\mathcal{B}_R|)$ satisfies $u(w) \leq 3u(0)$ on $|w| \leq R$: use the Poisson kernel on $R < s < 2R$, with ratio at most $(s+R)/(s-R)$, then let $s \uparrow 2R$. Therefore $|b/\mathcal{B}_R| \geq d_A^3/M_b^2$. Boundary zeros are allowed in this argument.

On $|w| = 2R$, $|\beta_\alpha - c'| \leq 4 + kR_h$, and the n_E actual strip entries give

$$|e^{-c'w} F_z/w^v| \leq \sqrt{n_E} e^{2(4+kR_h)R} \|z\|.$$

The numerator maximum principle and the preceding denominator bound prove

$$\sup_{|w| \leq R} |A_z(w)| \leq M_k(R) \|z\|, \quad z \in \mathcal{Z}_R. \quad (\text{RG24})$$

The row coefficient norm is exactly the one induced by the fixed R_Z congruence.

RG9. Every allowed growing radius and its complementary cost

Let $L_k \rightarrow \infty$, $L_k = o(k)$, and $R_k = k/L_k$. For the complete metric-and-sampling result use the explicit finite guards

$$1 \leq L_k \leq k/4, \quad R_k = k/L_k, \quad a_0 + b_0 R_k \leq r/2, \quad (\text{RG25})$$

together with (RG1) and the original finite arithmetic-receiver guards when using that receiver. All radius guards hold eventually for fixed data. The actual codimension $h = h_{R_k}$ is an integer, $s = r - h \geq r/2 > 0$, and one coefficient-orthonormal inclusion $J_R : \mathbb{C}^s \hookrightarrow \mathbb{C}^r$ is fixed at every cutoff.

For $l_i = q - 1 + i$, $u_i = 2q - 1 + i$, $i = 0, 1$, put $G_i = G(l_i, u_i)$, $G_i^R = G_{J_R}(l_i, u_i)$. Then

$$0 \leq (G_0 + G_1) - (G_0^R + G_1^R) \leq 2\mathfrak{D}_k(h) = O\left(kq \frac{1 + \log L_k}{L_k}\right) = o(kq). \quad (\text{RG26})$$

Indeed $k/L_k \rightarrow \infty$ absorbs the fixed a_0 , so $h/r = O(1/L_k)$, and (RG15) applies. The entire attained complementary determinant is included.

The first arrival's choice is also valid: $d = \lfloor r/\log(q+e) \rfloor$, $R = (d - a_0)/(b_0 + 1)$, with finite guards $2 \leq R \leq k$, $R \geq \frac{1}{2} \log(8q - 1)$, $d \leq r/2$. Since $a_0 + b_0 R = d - R \leq d$, its true codimension is at most d ; all guards hold eventually. Its complementary cost is $O(kq \log \log(q+e)/\log(q+e))$. For sampling at this radius add $R \geq 4$, also eventual.

RG10. Finite-radius defect and repaired every-direction metric theorem

The conditions $L_k \rightarrow \infty$, $L_k = o(k)$ do **not** imply $R_k \geq \text{arctanh}(1 - 1/(4q)) = \frac{1}{2} \log(8q - 1)$. For example $L_k = k/\sqrt{\log(q+e)}$ satisfies both conditions but has $R_k = \sqrt{\log(q+e)}$, smaller than this logarithmic threshold eventually. Thus the fixed t -circle of the first arrival cannot be used automatically for every later L_k . The exact admissible circle space is

$$\mathcal{D}_{k,R} = \{\rho : 0 < \rho \leq 1 - 1/(4q), \text{ arctanh } \rho \leq R\}.$$

Its largest member is

$$\rho_{k,R} = \min\{1 - 1/(4q), \tanh R\}. \quad (\text{RG27})$$

For $|t| \leq \rho_{k,R}$, the power series gives $|-i \arctan t| \leq \text{arctanh } |t| \leq R$. The original Gamma generating function yields

$$H_z(t) = (1 + t^2)^{-1/4} A_z(-i \arctan t), \quad W_{z,n} = \frac{\rho_n}{\sqrt{M_\sigma}} [t^n] H_z, \quad \rho_n^2 = \frac{n!}{(1/2)_n} \leq 2n + 1.$$

The last inequality follows by induction. Cauchy's bound and $\sum_{n=0}^D (2n+1) = (D+1)^2$, applied to every retained row combination, give

$$b_k^- I \preceq B_N^R := J_R^* B_N J_R \preceq b_{k,R}^+ I, \quad b_{k,R}^+ = \max\{1, M_k(R)^2 M_\sigma^{-1} (1 - \rho_{k,R}^2)^{-1/2} (2q+1)^2 \rho_{k,R}^{-4q}\}. \quad (\text{RG28})$$

There is no additional row-rank factor. A completely finite logarithmic bound is

$$\log b_{k,R}^+ \leq \log^+(M_k(R)^2/M_\sigma) + 2 \log(2q+1) + \frac{1}{2} \log(4q) + \frac{4}{3} + \frac{8qe^{-2R}}{1 - e^{-2R}}. \quad (\text{RG29})$$

Indeed $1 - \rho_{k,R}^2 \geq 1/(4q)$, $-4q \log(1 - 1/(4q)) \leq 4/3$, and

$$-\log \tanh R = \log \frac{1 + e^{-2R}}{1 - e^{-2R}} \leq \frac{2e^{-2R}}{1 - e^{-2R}}.$$

The negative logarithm of the minimum is the maximum of the two negative logarithms, bounded by their sum. Taking a positive part in (RG28) proves (RG29).

Define the finite nonnegative width

$$\epsilon_{k,R}^{\text{met}} = \frac{1}{q} \max\{\log^+(1/b_k^-), \log b_{k,R}^+\}. \quad (\text{RG30})$$

At $R = k/L_k$,

$$\epsilon_{k,R}^{\text{met}} = O\left(\frac{1}{L_k} + e^{-2k/L_k} + \frac{k \log(q+2)}{q}\right) = o(1). \quad (\text{RG31})$$

This repairs the general L_k extension in every retained direction. When $R \geq \frac{1}{2} \log(8q-1)$, the original fixed circle is valid and the exponential term is unnecessary; this holds eventually for $L_k = \sqrt{\log(q+e)}$ and the first arrival's radius.

RG11. Uniform Wasserstein comparison with the original angles

Let $y_{1,N} \geq \dots \geq y_{s,N}$ be the centered log eigenvalues of $(B_N^R)^{-1/2} C_N^R (B_N^R)^{-1/2}$, where $C_N^R = J_R^* C_N J_R$, and let $z_{j,N} = q^{-1} \log \lambda_j(C_N^R) + 2\ell_k$. The exact isometry

$$\mathcal{J}_N = B_N^{1/2} J_R (J_R^* B_N J_R)^{-1/2}$$

compresses the full angle matrix to the retained one. Thus

$$x_{j,N} \geq y_{j,N} \geq x_{j+h,N}, \quad |z_{j,N} - y_{j,N}| \leq \epsilon_{k,R}^{\text{met}}. \quad (\text{RG32})$$

The second statement is generalized min-max applied to (RG28).

Put $\mu_N = r^{-1} \sum \delta_{x_{j,N}}$, $\nu_N^R = s^{-1} \sum \delta_{z_{j,N}}$, $\delta_R = h/r$. At $h = 0$ interpret $\delta_R \log(1/\delta_R) = 0$. Then

$$\sup_N W_1(\mu_N, \nu_N^R) \leq \epsilon_{k,R}^{\text{met}} + \delta_R [4C_{\text{ang}} + 6 + 2 \log(1/\delta_R)]. \quad (\text{RG33})$$

Here is the full proof with both empirical masses. Pad $y_1, \dots, y_s$ by h copies of $-C_{\text{ang}}$. The resulting decreasing list $\tilde{y}$ satisfies $x_j \geq \tilde{y}_j$, by (RG9),(RG32). Its ordered-coupling cost is therefore its mean difference, bounded by

$$\frac{1}{r} \left(2hC_{\text{ang}} + 2 \sum_{j=1}^h \log(r/j) \right) \leq \delta_R [2C_{\text{ang}} + 2 + 2 \log(1/\delta_R)].$$

The padded law is $(1 - \delta_R)s^{-1} \sum \delta_{y_j} + \delta_R \delta_{-C_{\text{ang}}}$. Moving its additional atom to the restricted list costs at most

$$\delta_R \left(s^{-1} \sum y_j + C_{\text{ang}} \right) \leq \delta_R [2C_{\text{ang}} + 2 + 2 \log(r/s)].$$

Use $y_j \leq C_{\text{ang}} + 2 \log(r/j)$ and $s^{-1} \sum_{j \leq s} \log(s/j) \leq 1$. Since $\delta_R \leq 1/2$, $2 \log(r/s) \leq 2 \log 2 < 2$. The triangle inequality and (RG32) prove (RG33). Ordered coupling is optimal on the line: uncrossing $a \leq b, c \leq d$ uses $|a - c| + |b - d| \leq |a - d| + |b - c|$.

The right side increases when δ_R is replaced by any certified upper bound in $[\delta_R, 1/2]$, because the derivative of its last term is $4C_{\text{ang}} + 4 + 2 \log(1/\delta) > 0$. At the general radius it is

$$O\left(\frac{1 + \log L_k}{L_k} + e^{-2k/L_k} + \frac{k \log(q+2)}{q}\right) = o(1). \quad (\text{RG34})$$

Every 1-Lipschitz statistic, including the mean and clipped sums, differs by at most (RG33). Two endpoint differences have at most twice that error. Since C_N^R increases,

each $z_{j,N}$ increases: ν_N^R is an exact stochastically increasing path approximating the full angular path uniformly. Its total W_1 variation telescopes and is at most

$$\frac{G_J(q-1, 2q)}{sq} \leq \frac{\mathfrak{D}_k(r)}{sq} \leq \frac{r}{s}(B_{\text{inc}} + 2\alpha). \quad (\text{RG35})$$

Uniform path distance alone is not used to infer total variation for the original path.

Set $C_R = C_{\text{ang}} + \epsilon_{k,R}^{\text{met}} + 2\log(r/s)$. Then $-C_R \leq z_j \leq C_R + 2\log(s/j)$. Therefore $s^{-1} \sum (z_j - T)_+ \leq 2e^{-(T-C_R)/2}$ for $T \geq C_R$, and $s^{-1} \sum e^{\theta z_j} \leq e^{\theta C_R}/(1-2\theta)$ for $0 \leq \theta < 1/2$. The proof is the same index count and integral as RG4. Uniform first-moment control, including its tails, is retained.

RG12. Both original Gamma source orders

Write $k = 4\ell + 1$. Retain the full masses

$$dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad s \in \{1, k\},$$

$$dm_k = \beta_k \prod_{j < \ell} [y^2 + (2j + 1/2)^2] d\sigma, \quad \beta_k = \frac{(2\pi)^{k/2}}{\sqrt{2}\Gamma(k/2)}. \quad (\text{RG36})$$

The equality is the Gamma recurrence applied exactly ℓ times. The orthonormal polynomial recurrence implies $\|yp\|_\sigma \leq 2(D+1)\|p\|_\sigma$ for degree at most D : the two off-diagonal weighted shifts each have norm at most $D+1$. Apply this successively to multiplication by $y + i(2j + 1/2)$. With $D_{\text{max}} = 2q - v$, set

$$a = \beta_k \prod_{j < \ell} (2j + 1/2)^2, \quad b = \beta_k \prod_{j < \ell} [2(D_{\text{max}} + j + 1) + 2j + 1/2]^2, \quad \Lambda_\Gamma = \log(b/a). \quad (\text{RG37})$$

The complete source and every complete ideal satisfy $aH^{(1)} \preceq H^{(k)} \preceq bH^{(1)}$. The exact dual supremum reverses these inequalities for C, B , including their row restrictions. Hence every original angle logarithm changes by at most Λ_Γ , and the centered laws differ by at most $\Lambda_\Gamma/q = O(k \log q/q) = o(1)$.

The same-source relative-growth and sampling constants transport exactly as

$$c^{(k)} = c_k/b, \quad b_-^{(k)} = b_k^-/b, \quad \mathfrak{U}^{(k)} = \mathfrak{U}_k/a, \quad H_{k,R}^{(k)} = H_{k,R}/\sqrt{a}. \quad (\text{RG38})$$

Thus $\Omega^{(k)} = \Omega + \Lambda_\Gamma$; the additional rank- h exterior allowance is $h\Lambda_\Gamma$. The sampling threshold increases by at most $\Lambda_\Gamma/(2\log 2)$, plus its integer ceiling. The full β_k is retained in all form comparisons. If projected-covariance laws themselves are compared between sources, the common mass shift must also be kept: corresponding log eigenvalues lie between shifts $-q^{-1} \log b$ and $-q^{-1} \log a$. The angle comparison cancels that common scale only after forming the ratio of both physical forms.

RG13. Full-ideal estimate for arbitrary finite coefficient errors

This argument is restated to cover sampling errors that need not satisfy any pole-cancellation equations. The original Gamma product implies

$$\sigma(y) \geq c_\sigma e^{-\pi|y|/2} / (1 + 4y^2).$$

Indeed retain its first factor and bound the remaining sum of logarithms by $\int_0^\infty \log(1 + t^2/x^2) dx = \pi|t|$, at $t = y/2$. On the original interval $[q, 2q]$, all lower roots have modulus at most $q/2$, so

$$\|Q - p\|_\sigma^2 \geq \frac{c_\sigma q e^{-\pi q}}{1 + 16q^2} (q/2)^{2q'} \|p(q \cdot)\|_{L^2[1,2]}^2.$$

The norm on the left is the full real-line norm. In the shifted orthonormal Legendre frame, the coefficient sum of the degree- j polynomial is $\sqrt{2j+1}P_j(5) \leq \sqrt{2j+1}10^j$. All its roots lie in $(1, 2)$ by orthogonality and sign changes, so its coefficients alternate and evaluation at -1 gives this sum. The bound on $P_j(5)$ follows from

$$P_j(x) = \pi^{-1} \int_0^\pi (x + \sqrt{x^2 - 1} \cos t)^j dt,$$

whose summed generating series is $(1 - 2xz + z^2)^{-1/2}$. Cauchy-Schwarz across degrees at most M yields

$$\|\text{coef } p(q \cdot)\|_1 \leq (M+1)10^M \|p(q \cdot)\|_{L^2[1,2]} \leq \mathfrak{A} q^{-q'} \|Q_{-p}\|_\sigma. \quad (\text{RG39})$$

Let an arbitrary finite error functional e , through degree D , satisfy $|e[y^n]| \leq Hn!R^{-n}$. Since $n! \leq D^n$, its coefficient pairing with Q_{-p} is bounded by the product of absolute coefficient sums at $t = D/R$. All q' roots occur in $\sum |[y^j]Q_{-}|t^j \leq (kR_h + t)^{q'}$, and $\sum |p_j|t^j \leq \max(1, t/q)^M \sum |p_j|q^j$. Therefore

$$\|e|_{I_N}\| \leq \mathfrak{A} H(kR_h/q + D/(qR))^{q'} \max(1, D/(qR))^{D-q'}. \quad (\text{RG40})$$

Only the finite envelope was used: no holomorphic continuation or cancellation property of the errors was assumed.

RG14. Common-circle sampling in the original w -plane

For $R \geq 4$, write $A_z(w) = \sum a_{z,n} w^n$, choose one integer $T > D_{\max}$, and sample at $w_j = (R/2)e^{2\pi i j/T}$. Define

$$a_{z,n}^{[T]} = \frac{(R/2)^{-n}}{T} \sum_{j=0}^{T-1} A_z(w_j) e^{-2\pi i j n/T}, \quad 0 \leq n < T.$$

The finite root-of-unity sum and absolute convergence give

$$a_{z,n}^{[T]} - a_{z,n} = \sum_{m \geq 1} a_{z,n+mT} (R/2)^{mT}, \quad |a_{z,n}^{[T]} - a_{z,n}| \leq M_k(R) R^{-n} \frac{2^{-T}}{1 - 2^{-T}} \|z\|. \quad (\text{RG41})$$

The bound holds on every combination of retained orthonormal rows. Keep the complete derivative phase and apply (RG40); $D \leq 2q$, $R \geq 4$ make its last maximum one. The error on the whole lower ideal is

$$\|\widehat{\mathcal{R}}_{I,N} - \mathcal{R}_{I,N}\| \leq H_{k,R} \frac{2^{-T}}{1 - 2^{-T}}, \quad H_{k,R} = \mathfrak{A} M_k(R) (kR_h/q + 2/R)^{q'}. \quad (\text{RG42})$$

These maps have source I_N in its full Gamma norm and target $\mathbb{C}^s$ in the retained coefficient frame.

To compute the whole ideal, put

$$\begin{aligned} \mathbf{H}_{I,N}[i, j] &= \int_{\mathbb{R}} y^{i+j} |Q_{-}(y)|^2 d\sigma(y), \\ F_N[z, j] &= \sum_{t=0}^{q'} [y^t] Q_{-} (-i)^{t+j} (t+j)! a_{z,t+j}, \quad 0 \leq i, j \leq M. \end{aligned}$$

Then exactly

$$C_N^R = F_N \mathbf{H}_{I,N}^{-1} F_N^*. \quad (\text{RG43})$$

Indeed multiplication by Q_- has an injective coefficient matrix J_I whose image is the full evaluation kernel. In orthonormal source coordinates its orthogonal projector is $J_I(J_I^* J_I)^{-1} J_I^* = P_N$. Expressing this identity in monomial coordinates gives (RG43). The mass M_σ , all phases and the physical $i^{q'}$ factor remain. A shared sampled coefficient prefix supplies all four cutoffs and every intermediate N . Exact nested Gram restrictions preserve the source nesting for these approximated functionals.

RG15. Relative perturbation, finite sample count and precision

Let $L = \mathcal{R}_{I,N}$, so $LL^* = C_N^R \succeq c_k I$. If $\|\hat{L} - L\| \leq \eta\sqrt{c_k}$, $0 < \eta < 1/2$, then every target vector satisfies

$$\|\|\hat{L}^* x\| - \|L^* x\|\| \leq \eta\sqrt{c_k}\|x\| \leq \eta\|L^* x\|.$$

Squaring proves

$$(1 - \eta)^2 C_N^R \preceq \hat{C}_N^R \preceq (1 + \eta)^2 C_N^R. \quad (\text{RG44})$$

No second condition-number factor is needed.

Since $s > 0$, applying the true functional bound to a retained unit row gives $H_{k,R} \geq \sqrt{c_k}$. A sufficient integer count is

$$T = \max \left\{ 2q - v + 1, \left\lceil \log_2 \frac{2H_{k,R}}{\eta\sqrt{c_k}} \right\rceil \right\}. \quad (\text{RG45})$$

It gives $2^{-T} \leq \eta\sqrt{c_k}/(2H_{k,R}) \leq \eta/2 < 1/4$, so (RG42) is at most $\eta\sqrt{c_k}$. The degree guard and ceiling remain explicit.

At $R = k/L_k$,

$$\log \frac{H_{k,R}}{\sqrt{c_k}} = q' \log L_k + O(q), \quad T = \frac{q'}{\log 2} \log L_k + O(q + \log(1/\eta)). \quad (\text{RG46})$$

For the first equality, write $kR_h/q + 2/R = (L_k/k)(2 + k^2 R_h/(qL_k))$; its final factor has a bounded positive logarithm on $1 \leq L_k \leq k/4$. Combine with $\frac{1}{2} \log(1/c_k) = (q - 1) \log(q/k) + O(q)$, $\log \mathfrak{A} = O(q)$, and $\log M_k(R) = O(q/L_k + \log q)$. The remaining $\log k$ term is $(q - 1 - q') \log k = (16k - 49) \log k = O(q)$; also $(q - 1) \log(q/k^2) = O(k)$. This proves the stated leading coefficient. Eventually $\log L_k \rightarrow \infty$, so the degree guard contributes only $O(q)$. In particular,

$$L_k = \sqrt{\log(q + e)} \implies T = \frac{q'}{2 \log 2} \log \log(q + e) + O(q + \log(1/\eta)). \quad (\text{RG47})$$

This is T sample points per retained row, or T vector evaluations; it uses sT scalar values. It is distinct from CS1-24's total $O(kq)$ complete-basis theorem, which uses different radii and retains every complementary row.

For $\mathcal{R}f = f_{q-1} + f_q - f_{2q-1} - f_{2q}$, the per-cutoff determinant errors in (RG44) lie between $2s \log(1 - \eta)$ and $2s \log(1 + \eta)$. Hence

$$|\mathcal{R} \log \det \hat{C}_N^R - \mathcal{R} \log \det C_N^R| \leq 4s \log \frac{1 + \eta}{1 - \eta} \leq 8s[-\log(1 - \eta)]. \quad (\text{RG48})$$

For any $\varepsilon > 0$, use $\eta = \min\{1/4, 1 - e^{-\varepsilon/(8s)}\}$. Then (RG48) is at most ε , and the remainder in (RG47) is $O(q + \log^+(s/\varepsilon))$.

If each sampled row vector has Euclidean error at most δ_{eval} , the coefficient envelope is $\delta_{\text{eval}}(R/2)^{-n}$ on every unit row combination. Equation (RG40) bounds the additional map error by

$$\mathfrak{A}(kR_h/q + 4/R)^{q'} \delta_{\text{eval}}. \quad (\text{RG49})$$

Add this to (RG42) before applying (RG44). Per-row scalar error δ gives $\delta_{\text{eval}} \leq \sqrt{s}\delta$. If the full source Gram obeys $(1-\tau)H \preceq \hat{H} \preceq (1+\tau)H$, $0 \leq \tau < 1$, its complete ideal has the same enclosure. Inversion changes the factors of (RG44) to $(1-\eta)^2/(1+\tau)$ and $(1+\eta)^2/(1-\tau)$. Errors in roots, periods, cancellation jets and row construction must enter these actual function, coefficient and Gram budgets, as specified in CS21–24. At canceled zeros, use the removable analytic quotient; a rounded $0/0$ is not a valid sample.

RG16. Full original arithmetic-kernel receiver

Keep $g = q - q' - v$ and $\mathcal{K}_k = \mathcal{R} \log \det(I_K^* G_N I_K)$. The exact finite scalar is

$$\nu_{2j} = (2j)! [t^{2j}] (\cos t)^{-1/2}, \quad \nu_{2j+1} = 0, \quad \Delta_n(a) = \det[\nu_{2a+i+j}]_{i,j=0}^n, \quad \Delta_{-1} = 1,$$

$$\mathcal{A}_{k,v} = \frac{\Delta_{g-1}(q') \Delta_g(q') \Delta_{q-1}(q-v) \Delta_q(q-v)}{\Delta_{q+g-1}(q') \Delta_{q+g}(q') \Delta_0(q-v)}. \quad (\text{RG50})$$

The complete published finite proof FI1–13 proves on its original KF/CG finite guards

$$|\mathcal{K}_k - \log \mathcal{A}_{k,v} + G_0 + G_1| \leq \mathcal{E}_k, \quad (\text{RG51})$$

with its complete original error

$$\mathcal{E}_k = B_{\text{KF49}}^{\text{sharp}} + \sum_{N \in \{q-1, q, 2q-1, 2q\}} [E_N^{\text{vol}} + (M_N + 1)E_{\chi'} + (L_N + 1)E_{d,N}],$$

$$L_N = N - q, \quad M_N = L_N + g, \quad B_{\text{KF49}}^{\text{sharp}} = 2m\Lambda_k + 2s_k \log \kappa_k^* + 2B_k^{\text{ang}} + 2 \sum_N E_N^*, \quad m = 8k - 16,$$

$$\Lambda_k = \log(u_k / \ell_{k,2q}), \quad \ell_{k,2q} = a_k / [1 + 4(2q + 25)^2]^{25}, \quad u_k = A_k,$$

$$E_N^* = (N + 1) \log M_{N-v} + J_{N-v}, \quad J_D = (D + 1) \log^+(1/d_A), \quad B_k^{\text{ang}} = v \log \kappa_k^* + 2v \log(1/\epsilon_k). \quad (\text{RG52})$$

The constants $a_k, A_k, M_D, \kappa_k^*, \epsilon_k$, CG24's E_N^{vol} , and CG18–20's exact $E_{\chi'}, E_{d,N}$ are the original quantities fully defined and proved in FI and its retained provider sources. They concern the arithmetic source, the complete polynomial factors and both original integration regions; they are not replaced by (RG4)'s coefficient-frame constants. FI gives $\Lambda_k = O(k + \log q)$, $E_{\chi'} = O(1)$, $E_{d,N} = O(\log q)$, $\mathcal{E}_k = O(q \log(q + 2))$. This continuation changes none of those original source factors or guards. Equation (RG51) is a proved finite determinant identity, not an assumed evaluation of a profile coefficient.

The physical coefficient map $(c' + iy)^j = \sum_{a \leq j} \binom{j}{a} (c')^{j-a} i^a y^a$ has unit-modulus diagonal phases and is applied to the complete functional and Gram matrices. The full Gamma mass contributes the same dimension difference g at every cutoff; it cancels under the four signs only after those dimensions are retained. The conductor polynomial $d_{\text{cond}}(S')$ stays distinct from the scalar d_A , with its complete leading coefficient and root factors present in FI's Gram identity.

Define $\mathcal{K}_{k,R}^{\text{ret}} = \log \mathcal{A}_{k,v} - G_0^R - G_1^R$. Equations (RG20), (RG51) prove the directed finite interval

$$\mathcal{K}_{k,R}^{\text{ret}} - \mathcal{E}_k - 2\mathfrak{D}_k(h_R) \leq \mathcal{K}_k \leq \mathcal{K}_{k,R}^{\text{ret}} + \mathcal{E}_k. \quad (\text{RG53})$$

If sampling has four-sign determinant error at most ε , replace the retained receiver by its sampled value and enlarge each side's allowance by ε . The scalar $g = 16k - 48 - v$ remains unchanged; it is not the retained row rank s . At $L_k = \sqrt{\log(q + e)}$, the full allowance is

$$O\left(kq \frac{1 + \log \log(q + e)}{\sqrt{\log(q + e)}} + q \log(q + 2)\right) = o(kq). \quad (\text{RG54})$$

The original source-order comparison (RG36)–(RG38) adds its explicitly stated finite costs. The remaining projected covariance still contains every original lower-root constraint; its kq -coefficient has not been assigned a value.

RG17. Verification scope

The companion `check_growth.py` checks noncommuting complex forms, restriction and attained-quotient interlacing, complete complementary Schur factors, inverse cross terms, all selected chronological subsets of its fixture, full-root innovations, unequal empirical masses, a radius-guard counterexample, coefficient aliasing and relative map perturbation. `GROWTH_CHECKS.json` and `GROWTH_CHECKS_OPTIMIZED.json` record test counts, exact witnesses and source hashes. These finite auxiliary fixtures supplement the proofs; they do not evaluate the period-dependent original row data. Existing AS and CS source bytes are preserved.

Original human sources

R. A. Askey and R. Roy, DLMF Chapter 5, original Gamma product, Eq. 5.8.3, supply the product used in RG13. T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt, DLMF Chapter 18, supply the Legendre generating function, Eq. 18.12.11, the Meixner–Pollaczek recurrence, Eq. 18.22.8, and its generating function, Eq. 18.23.7. The complete original formula TeX files, version 1.2.8 in the retained receipt, were read; their hashes and receiving calculations are listed in `SOURCE_USE_LEDGER.json`. The new relative-growth, attained-minimum, chronological, radius and Wasserstein results are the present derivation, not assertions attributed to those human sources.

Part III

Sampling every original invariant row

Complete covariance sampling with a finite input-error certificate

22 September 2026. This derivation proves the complete-row sampling calculation in the original coefficient, Gamma and root coordinates. It also gives an arbitrary-accuracy sample count and an explicit perturbation bound for the full lower-root multiplication map. The native period-dependent covariance is not numerically evaluated here.

CS1. Original spaces and the exact functional

Retain the original simple-quartet five-orbit domain, $k \equiv 1 \pmod{4}$, $k \geq 17$, $q = (k+1)^2$, $q' = (k-7)^2$, $0 < \delta < 1/2$, $\gamma > 2$, $R_h = \sqrt{\delta^2 + \gamma^2}$, and the finite guard $q \geq 2kR_h$. The fixed conductor is

$$E_A(w) = \sum_{a,b=0}^8 a_{ab} e^{b_{ab}w}, \quad b_{ab} = 4 + (2a-8)\delta + i(2b-8)\gamma, \quad v = \text{ord}_0 E_A, \quad \mu_v = E_A^{(v)}(0) \neq 0. \quad (\text{CS1})$$

Set $c' = k/2 - 4$, $g = q - q' - v$, $r = 8k - 32 - v + s_k$, $s_k = \dim(K \cap \mathcal{P}_{<v})$, and $n_E = 16k - 48$. The original complete invariant row space after low-degree elimination is $V_k \subset \mathbb{C}^{n_E}$, with its actual strip embedding $S_k : \mathbb{C}^{n_E} \hookrightarrow \mathbb{C}^q$ given by zero insertion. A row z means the full-root row zS_k^T in an exponential sum. For its original upper roots β_α , put

$$F_z(w) = \sum_{\alpha=1}^q (zS_k^T)_\alpha e^{\beta_\alpha w}, \quad A_z(w) = e^{-c'w} F_z(w) / E_A(w), \quad \ell_z(y^n) = (-i)^n n! [w^n] A_z(w). \quad (\text{CS2})$$

The complete low-degree constraint implies $F_z^{(j)}(0) = 0$ for $j < v$; the quotient is removable at zero. The phase and center in CS2 follow from $\partial_w^n (e^{-c'w} G) = e^{-c'w} (\partial_w - c')^n G$ and $y = (S' - c')/i$. This is the unweighted functional in the physical y coordinate. On a weighted coefficient polynomial p , its ideal restriction is $\ell_z(Q_- p)$, with the full factor Q_- inserted once.

For $q-1 \leq N \leq 2q$ put $D = N - v$, $M = D - q'$ and

$$Q_-(y) = i^{-q'} \chi'(c' + iy) = \prod_{a,b=0}^{k-8} (y - \omega'_{ab}), \quad \omega'_{ab} = (2b - k + 8)\gamma - i(2a - k + 8)\delta, \quad I_N = Q_- \mathcal{P}_M \subset \mathcal{P}_D. \quad (\text{CS3})$$

Every lower root occurs, with its exact coordinate. The source measure is $d\sigma(y) = |\Gamma(1/4 + iy/2)|^2 dy / (2\pi)$ and has mass $M_\sigma = \sqrt{2\pi}$. Its monic polynomials satisfy

$$p_{n+1} = yp_n - n(n-1/2)p_{n-1}, \quad \gamma_n = \|p_n\|_\sigma^2 = M_\sigma n! (1/2)_n, \quad \sum_{n \geq 0} \frac{p_n(y)}{n!} t^n = (1+t^2)^{-1/4} e^{y \arctan t}. \quad (\text{CS4})$$

The original NIST author TeX for DLMF18.22.8 and 18.23.7 was read. The authors are T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt. CS4 is their Meixner–Pollaczek specialization with the displayed full mass.

For the original row matrix Z , the fixed row change $R_Z^{-1/2}$, $R_Z = ZZ^*$, gives a coefficient-orthonormal frame of V_k . This is a congruence on row Grams; it changes neither the source norm nor the ideal. In that frame let W_N evaluate every ℓ_z on $p_n/\sqrt{\gamma_n}$. Let E_D evaluate those polynomials at every lower root. Interpolation at the q' distinct roots proves $\ker E_D = I_N$ and $\text{rank } E_D = q'$. Thus

$$P_N = I - E_D^*(E_D E_D^*)^{-1} E_D, \quad C_N = W_N P_N W_N^*, \quad B_N = W_N W_N^*. \quad (\text{CS5})$$

These denote the row-congruent original matrices throughout. The inverse congruence restores the physical row frame after calculation. If $r = 0$, all row determinants are one and the sample count is zero; below $r > 0$.

CS2. Analytic inputs with their exact source status

The complete independent original-angle source, AS6-26 of the complete original-angle proof, supplies the following finite estimates; its mathematical argument was read directly. These are proved analytic inputs, not consequences of its auxiliary checker. Write $A_1 = \sum |a_{ab}|$, $d_A = |\mu_v|/v!$, $B_A = \max_{a_{ab} \neq 0} |b_{ab}|$, $C_{\text{path}} = A_1/|a_*|$, where a_* is the actual nonzero lexicographic pivot. Define

$$\begin{aligned} \mathfrak{M}_k &= \max\{1, e^{1+2\pi\gamma} A_1 (8q+1)^{1/4} (2q+1)(4q+1)^{4\delta}\}, \\ B_{\text{val},k} &= q M_\sigma e^{8q} \left(\frac{2q + k R_h}{2\delta k} \right)^{2(q-1)}, \quad B_E = n_E M_\sigma \left(\frac{2n_E + k R_h}{2\delta} \right)^{2(n_E-1)}, \\ c_k &= (\mathfrak{M}_k^2 B_{\text{val},k} q C_{\text{path}}^{20k})^{-1}, \quad b_k^- = (\mathfrak{M}_k^2 B_E)^{-1}, \quad C_N \succeq c_k I, \quad B_N \succeq b_k^- I. \end{aligned} \quad (\text{CS6})$$

AS6 retains every upper-root gap: shells of radius j have at most $8j$ nodes, giving $\prod_{\beta \neq \alpha} |\omega_\alpha - \omega_\beta| \geq (2\delta k)^{q-1} e^{-4q}$. Full Lagrange interpolation gives $B_{\text{val},k}$. The exact conductor matrix has norm at most $\mathfrak{M}_k$; the full strip reduction has norm at most $\sqrt{q} C_{\text{path}}^{10k}$. Applying these maps to $W_z - b E_D$ and minimizing over every lower row b proves the first bound of CS6 after projection. Strip interpolation gives the second. In particular

$$\log(1/c_k) = 2(q-1) \log(q/k) + O_{h,A}(q), \quad \log(1/b_k^-) = O_{h,A}(k \log(q+2)). \quad (\text{CS7})$$

The exact first remainder is $8q + 2(q-1) \log((2 + k R_h/q)/(2\delta)) + 2 \log \mathfrak{M}_k + \log(q^2 M_\sigma) + 20k \log C_{\text{path}}$, so CS7 introduces no hidden $q \log k$ term.

For $R \geq 1$, let $\mathcal{Z}_R \subset V_k$ impose all numerator value-and-jet cancellations, through the full multiplicity of each zero of $E_A(w)/w^v$ in $|w| < 2R$. Set

$$a_0 = \log^+(A_1/d_A)/\log 2, \quad b_0 = 4B_A/\log 2, \quad C_A = \max(A_1, d_A)^2/d_A^3.$$

AS12-16 proves

$$\begin{aligned} \text{codim } \mathcal{Z}_R &\leq a_0 + b_0 R, \quad \max_{|w| \leq R} |A_z(w)| \leq M_k(R) \|z\|, \\ M_k(R) &= C_A \sqrt{n_E} e^{(2k R_h + 8 + 4B_A)R}, \quad z \in \mathcal{Z}_R. \end{aligned} \quad (\text{CS8})$$

The count uses the full entire function E_A/w^v . The quotient estimate divides numerator and denominator by the same radius- $2R$ finite Blaschke product, then applies the maximum principle and the Harnack factor three on $|w| \leq R$. Thus it bounds the removable values as well as values away from zeros. No independence of cancellation equations enters. For arbitrary rows, with $\bar{B}_A = \max(1, B_A)$, the original fixed disk is

$$\begin{aligned} \rho_A &= \min\{1, (v+1)! d_A / (2A_1 \bar{B}_A^{v+1} e^{\bar{B}_A})\}, \\ M_{0,k} &= 2\sqrt{n_E} \rho_A^{-v} d_A^{-1} e^{(4+k R_h)\rho_A}, \quad \max_{|w| \leq \rho_A} |A_z(w)| \leq M_{0,k} \|z\|. \end{aligned} \quad (\text{CS9})$$

The analytic estimates CS6–9 are finite and do not use the monic-profile limit, the DCR asymptotic, or an assigned native determinant coefficient. Their complete proof source accompanies this derivation in the retained programme collection. The later asymptotic ACC/DCR receiver has its separate source status and is unnecessary for the finite rational receiver in CS24 below.

CS3. Full-source and complete-ideal envelope estimates

Let a finite functional e on $\mathcal{P}_D$ satisfy $|e(y^n)| \leq Hn!R^{-n}$ for $0 \leq n \leq D$. This includes a moment error; it need not itself come from an exponential quotient. Extend its moment sequence by zero above D . For $t_R = \min(1/2, R/4)$, the power series gives $|\arctan t| \leq \operatorname{arctanh} t_R \leq R/2$ on $|t| = t_R$. Therefore

$$\left| (1+t^2)^{-1/4} \sum_{n=0}^D \frac{e(y^n)}{n!} (\arctan t)^n \right| \leq 2H(1-t_R^2)^{-1/4}.$$

Its coefficient through degree D is $e(p_n)/n!$, by CS4. Cauchy's estimate and $\rho_n^2 = n!/(1/2)_n \leq 2n+1$ show

$$\|e\|_{\mathcal{P}_D^*}^2 = \sum_{n=0}^D \frac{|e(p_n)|^2}{\gamma_n} \leq \frac{4H^2}{M_\sigma \sqrt{1-t_R^2}} \sum_{n=0}^D (2n+1)t_R^{-2n} \leq H^2 P_D(R)^2,$$

$$P_D(R) = \frac{2(D+1)t_R^{-D}}{\sqrt{M_\sigma}(1-t_R^2)^{1/4}}. \quad (\text{CS10})$$

The last step uses $t_R \leq 1$ and $\sum_{n=0}^D (2n+1) = (D+1)^2$. The bound on ρ_n follows inductively since $\rho_{n+1}^2/\rho_n^2 = (n+1)/(n+1/2)$ and $(2n+1)(n+1)/(n+1/2) = 2n+2 \leq 2n+3$.

Here is the complete ideal estimate, including the original physical norm. The original Gamma-product TeX, DLMF5.8.3, by R. A. Askey and R. Roy, gives

$$\sigma(y) \geq \frac{c_\sigma e^{-\pi|y|/2}}{1+4y^2}, \quad c_\sigma = \Gamma(1/4)^2/(2\pi).$$

Indeed isolate the $n=0$ product factor and bound the remaining logarithmic sum by $\int_0^\infty \log(1+t^2/x^2)dx = \pi|t|$, with $t=y/2$. Differentiating that integral in $|t| > 0$ gives π and its limit at zero is zero. On $q \leq y \leq 2q$, every lower-root factor has modulus at least $q/2$, by the finite guard. Hence

$$\|Q_{-p}\|_\sigma^2 \geq \frac{c_\sigma q e^{-\pi q}}{1+16q^2} (q/2)^{2q'} \|p(q \cdot)\|_{L^2[1,2]}^2.$$

Expanding $p(qx)$ in the orthonormal Legendre polynomials $\sqrt{2n+1}P_n(2x-3)$ gives $\|\operatorname{coef} p(q \cdot)\|_1 \leq (M+1)10^M \|p(q \cdot)\|_2$. To verify the coefficient bound, the n roots of $P_n(2x-3)$ lie in $(1, 2)$ by its orthogonality and the sign-change argument: multiplying its interior sign-change factors would otherwise be a lower-degree polynomial with nonzero integral against it. Its coefficients therefore alternate, so their absolute sum is $P_n(5)$. The integral representation obtained by summing the original Legendre generating function, DLMF18.12.11 gives $P_n(5) \leq (5+\sqrt{24})^n \leq 10^n$. Cauchy-Schwarz and $\sum_{n \leq M} (2n+1) = (M+1)^2$ give the stated coefficient bound.

Put $M_{\max} = 2q - v - q'$ and

$$\mathfrak{A}_k = (M_{\max} + 1)10^{M_{\max}} 2^{q'} e^{\pi q/2} \sqrt{(1+16q^2)/(c_\sigma q)}.$$

Then $\sum_j |p_j| q^j \leq \mathfrak{A}_k q^{-q'} \|Q_-\|_\sigma$. As $n! \leq D^n$ for $n \leq D$, put $t = D/R$. The weighted absolute coefficient sum of the full root polynomial is at most $(kR_h + t)^{q'}$, and $\sum |p_j| t^j \leq \max(1, t/q)^M \sum |p_j| q^j$. Consequently

$$\|e|_{I_N}\| \leq H \mathfrak{A}_k (kR_h/q + D/(qR))^{q'} \max(1, D/(qR))^{D-q'}. \quad (\text{CS11})$$

All coefficients of Q_- were included before taking absolute values. This proof applies identically to the original functional and to any finite error sequence with the envelope in CS10.

For $2 \leq R \leq k$, CS8 and CS11 give $\|\ell_z|_{I_N}\| \leq \mathfrak{F}_k R^{-q'} \|z\|$, where

$$\mathfrak{F}_k = \mathfrak{A}_k C_A \sqrt{n_E} e^{2R_h q + (8+4B_A)k} (R_h + 2)^{q'}.$$

For all rows CS9 gives $\|\ell_z|_{I_N}\| \leq \mathfrak{F}_{0,k} \|z\|$, with

$$\mathfrak{F}_{0,k} = \mathfrak{A}_k M_{0,k} (kR_h/q + 2/\rho_A)^{q'} (2/\rho_A)^{M_{\max}}. \quad (\text{CS12})$$

Their logarithms are $O_{h,A}(q)$, since the original conductor is fixed, $M_{\max} \leq 2q$, $q' \leq q$, and $k^2 \leq q$.

CS4. One complete basis uses the nested cancellation spaces

Set $L_A = b_0 + 9$, $j_A = \lceil \max\{2(a_0 + 1), a_0 + 1 + 4L_A\} \rceil$, and $R_j = (j - 1 - a_0)/(2L_A)$ for $j \geq j_A$. Then $R_j \geq 2$, $R_j \geq j/(4L_A)$ and $R_j < r/(2L_A) \leq (8k - 32)/18 < k$. Moreover $\text{codim } \mathcal{Z}_{R_j} \leq a_0 + b_0 R_j \leq (j - 1 + a_0)/2 < j$. Choose $z_r, z_{r-1}, \dots, z_{j_A}$ successively. At stage j , all previously chosen vectors belong to $\mathcal{Z}_{R_j}$, since its radius is smaller, and their span has dimension $r - j$. The space $\mathcal{Z}_{R_j}$ has dimension at least $r - j + 1$, so its intersection with the orthogonal complement of that span has positive dimension. Select a unit vector there. Complete the early part by any orthonormal basis of the remaining orthogonal complement. If $j_A > r$, simply choose a complete orthonormal basis. This constructs a fixed unitary row map U and retains exactly r rows at every cutoff.

Define $\mathcal{R}_j = \rho_A$, $\mathcal{M}_j = M_{0,k}$ for early rows, and $\mathcal{R}_j = R_j$, $\mathcal{M}_j = M_k(R_j)$ for late rows. Put

$$\mathfrak{U}_k = \max\{\mathfrak{F}_k^2 (4L_A)^{2q'}, \mathfrak{F}_{0,k}^2 j_A^{2q'}\}, \quad B_j = \sqrt{\mathfrak{U}_k} j^{-q'}, \quad F_j = \mathcal{M}_j P_{2q}(\mathcal{R}_j). \quad (\text{CS13})$$

CS10–12 prove $\|\ell_{z_j}|_{I_N}\| \leq B_j$ and $\|\ell_{z_j}\|_{\mathcal{P}_D^*} \leq F_j$. They also prove the same bounds multiplied by ε for every error sequence whose Taylor coefficients obey $|e_n| \leq \varepsilon \mathcal{M}_j \mathcal{R}_j^{-n}$. This last assertion uses the finite envelope proof, not an assumption that an aliased sequence has the same pole cancellations. Uniformly in j , $\log^+ F_j = O(q)$: late radii satisfy $kR_j \leq k^2$, and early radii are fixed positive numbers. Since $C_N \succeq c_k I$, each unit row has ideal norm at least $\sqrt{c_k}$, so $B_j/\sqrt{c_k} \geq 1$; similarly $F_j/\sqrt{b_k^-} \geq 1$.

CS5. Exact aliasing and an arbitrary-accuracy sample count

Fix $0 < \eta \leq 1/8$ and set $a_j = \mathcal{R}_j/2$. For $T_j > 2q$ sample the exact removable function A_{z_j} at $a_j \exp(2\pi i \ell/T_j)$, $0 \leq \ell < T_j$. The Fourier coefficient estimate through degree $2q$ is

$$a_{j,n}^{[T]} = \frac{a_j^{-n}}{T_j} \sum_{\ell=0}^{T_j-1} A_{z_j}(a_j e^{2\pi i \ell/T_j}) e^{-2\pi i n \ell/T_j}.$$

The Taylor series converges absolutely at these points, since it is holomorphic on a neighborhood of the closed radius- $\mathcal{R}_j$ disk. Interchanging its sum with the finite Fourier sum gives, for $0 \leq n < T_j$,

$$a_{j,n}^{[T]} - [w^n]A_{z_j} = \sum_{h \geq 1} [w^{n+hT_j}]A_{z_j} a_j^{hT_j}, \quad |a_{j,n}^{[T]} - [w^n]A_{z_j}| \leq \frac{\mathcal{M}_j \mathcal{R}_j^{-n}}{2^{T_j} - 1}. \quad (\text{CS14})$$

The physical moments are $(-i)^n n! a_{j,n}^{[T]}$. Applying the finite-envelope bounds CS10-13 gives ideal row error $B_j/(2^{T_j} - 1)$ and full-source row error $F_j/(2^{T_j} - 1)$ before any Gram is formed.

Choose

$$T_j(\eta) = \max \left\{ 2q + 1, \left\lceil \log_2 \left[\frac{4\sqrt{r}}{\eta} \max \left(\frac{B_j}{\sqrt{c_k}}, \frac{F_j}{\sqrt{b_k^-}} \right) \right] \right\rceil \right\}. \quad (\text{CS15})$$

Let A_j denote the maximum in this formula; $A_j \geq 1$. Then

$$\frac{B_j}{2^{T_j} - 1} \leq \frac{\eta \sqrt{c_k}}{(4 - \eta)\sqrt{r}}, \quad \frac{F_j}{2^{T_j} - 1} \leq \frac{\eta \sqrt{b_k^-}}{(4 - \eta)\sqrt{r}}.$$

Permit an additional numerical row error at most $\eta \sqrt{c_k}/(2\sqrt{r})$ on the ideal and $\eta \sqrt{b_k^-}/(2\sqrt{r})$ on the full source. The Frobenius bound, namely the square root of the sum of squared row norms, bounds each operator error. Since $1/(4 - \eta) + 1/2 < 1$ on the stated range, the resulting map errors are strictly less than $\eta \sqrt{c_k}$ and $\eta \sqrt{b_k^-}$ respectively, simultaneously for all N .

Scalar sample errors can meet both budgets. A value error of modulus at most ϵ_j at every sample gives coefficient error at most $\epsilon_j a_j^{-n}$ by the finite Fourier sum. Define

$$D_j^I = \mathfrak{A}_k(kR_h/q + 2/a_j)^{q'} \max(1, 2/a_j)^{M_{\max}}.$$

Then the sufficient input enclosure is

$$\epsilon_j \leq \frac{\eta}{2\sqrt{r}} \min \left\{ \frac{\sqrt{c_k}}{D_j^I}, \frac{\sqrt{b_k^-}}{P_{2q}(a_j)} \right\}. \quad (\text{CS16})$$

The bounds include all Taylor coefficients, the factorials and the derivative phases. Errors in the actual row coefficients, physical period and removable quotient must be included in these enclosures. At a true canceled zero a raw rounded quotient is not the target value: one evaluates the holomorphic quotient using its exact cancellation identity and enclosed derivatives. An approximate cancellation equation alone is not a proof that a larger circle is admissible. Construction and certification of the actual cancellation flag are the stated input model of the sample-count theorem.

Put

$$H_k(\eta) = \max \left\{ 0, \log \frac{4\sqrt{r}\mathfrak{U}_k}{\eta r^{q'} \sqrt{c_k}}, \max_j \log \frac{4\sqrt{r}F_j}{\eta \sqrt{b_k^-}} \right\}.$$

Then $\log(4\sqrt{r}A_j/\eta) \leq H_k(\eta) + q' \log(r/j)$, so

$$\sum_{j=1}^r T_j(\eta) \leq r(2q + 2) + \frac{rH_k(\eta) + q'[r \log r - \log(r!)]}{\log 2} \leq r \left(2q + 2 + \frac{H_k(\eta) + q'}{\log 2} \right). \quad (\text{CS17})$$

The first inequality uses $\max(a, b) \leq a + b$ for nonnegative a, b and the ceiling bound. For the second, $\log(r!) = \sum_{j=1}^r \log j \geq \int_1^r \log x \, dx = r \log r - r + 1$.

Here is the exact cancellation needed to estimate H_k , rather than an extra logarithm charged to every row. The terms $\frac{1}{2} \log \mathfrak{U}_k$ and all full-source terms are $O(q)$. By CS7 the remaining term is

$$(q-1) \log(q/k) - q' \log r = (q-1-q') \log k + (q-1) \log(q/k^2) - q' \log(r/k).$$

The first is $O(k \log k)$ because $q-1-q' = 16k-49$; the second is $O(k)$ because $q/k^2 = (1+1/k)^2$; and the last is $O(q)$ because r/k is bounded above and below by positive constants for the stated fixed-conductor domain. Consequently

$$H_k(\eta) = O_{h,A}(q) + \log(1/\eta), \quad \boxed{\sum_j T_j(\eta) = O_{h,A}(kq + k \log(1/\eta))}. \quad (\text{CS18})$$

For fixed accuracy this is the asserted $O(kq)$ total scalar count. Individual early rows may require $O(q \log q)$ samples. The same coefficients through degree $2q$ supply all four original cutoffs, and in fact the whole intervening window. CS18 is a count of scalar row-function evaluations, not of zero isolation, exact rank decisions, bit operations, or matrix inversion.

CS6. Errors before Grams, with the exact complete ideal

Let H_D be the original Gamma Gram on $1, y, \dots, y^D$, and let L_N be the moment row matrix, with entries $(-i)^n n! [w^n] A_{z_j}$. Let $J_N : \mathbb{C}^{M+1} \hookrightarrow \mathbb{C}^{D+1}$ contain every coefficient column of multiplication by the complete monic Q_- . Then

$$B_N = L_N H_D^{-1} L_N^*, \quad C_N = (L_N J_N) (J_N^* H_D J_N)^{-1} (L_N J_N)^*. \quad (\text{CS19})$$

To verify equality with CS5, put $T = H_D^{1/2} J_N (J_N^* H_D J_N)^{-1/2}$. Its columns are orthonormal and span the full ideal in source orthonormal coordinates; hence TT^* is precisely P_N in those coordinates. The map $L_N H_D^{-1/2}$ is the full original functional matrix. Substitution proves CS19. The physical S' frame includes the factor $i^{q'}$ in J_N and its corresponding coefficient transform; both numerator and denominator of CS19 carry that same factor, so it cancels there exactly. No lower-root row has been removed.

For either full or ideal map write its original covariance as $X = KK^* \succeq xI$. If $\|\hat{K} - K\| \leq \eta\sqrt{x}$, then for every row-space vector u , $\|\hat{K}^*u - K^*u\| \leq \eta\|K^*u\|$. Both triangle inequalities, followed by squaring, give $(1-\eta)^2 X \preceq \hat{K} \hat{K}^* \preceq (1+\eta)^2 X$. This compares the maps before forming Grams and therefore does not multiply the error by the largest singular value of K .

If $(1-\tau)H_D \preceq \hat{H}_D \preceq (1+\tau)H_D$, $0 \leq \tau < 1$, the same inequalities hold after congruence by the complete J_N , and inverse order gives

$$\boxed{\alpha X_N \preceq \hat{X}_N \preceq \beta X_N \quad (X = C, B), \quad \alpha = \frac{(1-\eta)^2}{1+\tau}, \quad \beta = \frac{(1+\eta)^2}{1-\tau}.} \quad (\text{CS20})$$

For the received choices $\eta = 1/32$, $\tau = 1/16$, the stronger exact interval is $\alpha = 961/1088$, $\beta = 1089/960$, inside $[3/4, 5/4]$. This gives the same received conclusion with a smaller finite determinant error. The original Gamma moment matrix can be represented as $H_D = M_\sigma[\nu_{i+j}]$, where $\nu_j = j![t^j](\cos t)^{-1/2} \in \mathbb{Q}$. Thus exact rational moment arithmetic and a scalar enclosure of the physical mass alone provide the source-form enclosure; entrywise rounding of an ill-conditioned moment matrix is unnecessary. This representation retains M_σ explicitly. To prove the moment formula, integrate CS4 coefficient by coefficient: orthogonality makes the integral of p_n zero for $n > 0$. Substitute

$t = \tan u$ into that formal identity to obtain $\sum_{n \geq 0} u^n \int y^n d\sigma/n! = M_\sigma(\cos u)^{-1/2}$. Each coefficient uses finitely many moments, so this argument requires no interchange of an infinite integral and an analytic series.

CS7. A certificate when every lower-root coefficient is enclosed

The preceding received formula uses the exact matrix J_N . Here is an explicit extension for an approximated full multiplication matrix $\hat{J}_N$, with no omitted root. Write $G = J_N^* H_D J_N > 0$, $\Delta J = \hat{J}_N - J_N$, and

$$\delta_N = \|H_D^{1/2} \Delta J G^{-1/2}\| < 1, \quad F_* = \left(\sum_{j=1}^r F_j^2 \right)^{1/2}, \quad d_N = \frac{(F_* + \eta \sqrt{b_k^-}) \delta_N}{\sqrt{c_k}}.$$

The moment matrix $\hat{L}_N$ has the sampled map-error bounds proved above. Set $T = H_D^{1/2} J_N G^{-1/2}$, $\hat{T} = H_D^{1/2} \hat{J}_N G^{-1/2}$, so $T^* T = I$ and $\|\hat{T} - T\| = \delta_N$. Both triangle inequalities give $(1 - \delta_N)^2 I \preceq \hat{T}^* \hat{T} \preceq (1 + \delta_N)^2 I$. Also

$$\|\hat{L}_N \hat{J}_N G^{-1/2} - L_N J_N G^{-1/2}\| \leq \eta \sqrt{c_k} + (F_* + \eta \sqrt{b_k^-}) \delta_N = (\eta + d_N) \sqrt{c_k}.$$

The first term is the error on the exact ideal. The second uses the entire full-source map bound $\|\hat{L}_N H_D^{-1/2}\| \leq F_* + \eta \sqrt{b_k^-}$. Thus no unprojected norm is incorrectly substituted for the small ideal lower bound. Combining these two comparisons with the source error τ proves

$$\boxed{\frac{(1 - \eta - d_N)^2}{(1 + \tau)(1 + \delta_N)^2} C_N \preceq (\hat{L}_N \hat{J}_N)(\hat{J}_N^* \hat{H}_D \hat{J}_N)^{-1} (\hat{L}_N \hat{J}_N)^* \preceq \frac{(1 + \eta + d_N)^2}{(1 - \tau)(1 - \delta_N)^2} C_N,} \quad (\text{CS21})$$

where $\eta + d_N < 1$. In detail, express both the numerator and denominator in the fixed coefficient frame $G^{-1/2}$, apply CS20's vector comparison to the numerator, and inverse order to the denominator. There is no comparison of independently chosen square-root gauges. The middle matrix uses the complete polynomial with enclosed coefficients and approximates the exact original ideal by this displayed estimate; it is not asserted to equal that ideal's projection before the error is paid.

Every quantity in this condition has a finite coefficient enclosure. For example, choose certified $h_+ \geq \lambda_{\max}(H_D)$ and $g_- \leq \lambda_{\min}(G)$, both positive. Then

$$\delta_N \leq \sqrt{h_+/g_-} \|\Delta J\|_F \leq \sqrt{h_+/g_-} \sqrt{M+1} \|\text{coef}(\hat{Q}_- - Q_-)\|_2. \quad (\text{CS22})$$

The second inequality follows because every column of ΔJ is a shift of the full coefficient error vector. Bounds $h_+ = \text{tr } H_D$ and $g_- = \det G / (\text{tr } G)^M$ are valid: the product of the other M positive eigenvalues is at most $(\text{tr } G)^M$. Certified bounds on the positive trace and determinant give numerical versions, or interval elimination gives sharper ones. Positivity is proved independently from the injectivity of multiplication by the monic Q_- , so sufficient refinement of enclosures yields a positive margin.

If each of the q' original roots is enclosed to error at most ϵ_ω , and all exact roots have modulus at most $R_- = (k-8)R_h$, telescoping the two full products gives

$$\|\text{coef}(\hat{Q}_- - Q_-)\|_1 \leq q' \epsilon_\omega (1 + R_- + \epsilon_\omega)^{q'-1}.$$

Indeed replace the factors one at a time; the changed factor is a constant of modulus at most ϵ_ω , and coefficient ℓ^1 norms are submultiplicative. This estimate preserves

the degree, original number of factors and monic leading coefficient. It requires neither root deletion nor a favorable rank decision.

A concrete common budget is

$$\eta = 1/64, \quad \tau \leq 1/16, \quad \delta_N \leq \min \left\{ \frac{1}{64}, \frac{\sqrt{c_k}}{64(F_* + \sqrt{b_k^-}/64)} \right\} \quad \text{for every cutoff.} \quad (\text{CS23})$$

Then $d_N \leq 1/64$. Formula CS21 gives the explicit common interval $\alpha_J = (31/32)^2/[(17/16)(65/64)^2]$, $\beta_J = (33/32)^2/[(15/16)(63/64)^2]$. Direct rational comparison shows $\alpha_J > 3/4 + 1/32$ and $\beta_J < 5/4 - 1/32$. Consequently one may additionally allow final absolute arithmetic error at most $c_k/32$ in the computed ideal covariance and still retain $3C_N/4 \preceq \widehat{C}_N \preceq 5C_N/4$. The full covariance has the stronger CS20 interval with $\eta = 1/64$, and similarly permits output error $b_k^-/32$. Alternatively compute the finite products and inverses exactly on Gaussian-rational centers. The extra input coefficient precision in CS22 is displayed explicitly; CS18 does not claim to bound its bit cost.

CS8. Original determinant consequence and source transport

Suppose $\alpha C_N \preceq \widehat{C}_N \preceq \beta C_N$ at all four cutoffs, with fixed $0 < \alpha \leq \beta$. Every log determinant error lies in $[r \log \alpha, r \log \beta]$. Two signs are positive and two negative, so

$$|\mathcal{R} \log \det \widehat{C}_N - \mathcal{R} \log \det C_N| \leq 2r \log(\beta/\alpha).$$

Write $\mathcal{R}f_N = f_{q-1} + f_q - f_{2q-1} - f_{2q}$. The complete finite invariant-determinant proof, equations (2), (3), (12), (13) and FI1-4 of edition018, therefore gives

$$\left| \mathcal{K}_k - \log \mathcal{A}_{k,v} - \mathcal{R} \log \det \widehat{C}_N \right| \leq \mathcal{E}_k + 2r \log(\beta/\alpha). \quad (\text{CS24})$$

Here $\mathcal{K}_k = \mathcal{R} \log \det(I_K^* G_N I_K)$ is the original arithmetic kernel, $\mathcal{A}_{k,v}$ is the exact positive rational Hankel-determinant factor, and $\mathcal{E}_k$ is the complete physical comparison error with the same-source NG2 width and all original low, angle and exterior terms. All its finite guards remain. This receiver uses neither a monic-profile asymptotic nor the ACC/DCR replacement by gqC_∂ . In particular $[3/4, 5/4]$ gives only $2r \log(5/3)$ additional error in CS24; the sharper intervals of CS20-21 give their smaller corresponding constants. Fixed row congruences cancel in this four-sign expression because the same frame is used throughout.

If instead one uses the already proved ACC/DCR two-endpoint angle receiver, both C and B satisfy the same interval and the additional error in $2 \log(\det C_L \det B_H / (\det C_H \det B_L))$ is at most $4r \log(\beta/\alpha)$. Its analytic-profile status remains exactly that of the inherited receiver. This sampling argument does not establish that asymptotic or assign its unknown native coefficient.

For completeness, an actual proved full-source comparison $aH^\sigma \preceq H^\mu \preceq bH^\sigma$ on the same polynomial space transports every complete restriction and inverse by order. Thus the sampling constants become $c^\mu = c^\sigma/b$, $(b_-)^\mu = (b_-)^\sigma/b$, $B_j^\mu = B_j^\sigma/\sqrt{a}$, $F_j^\mu = F_j^\sigma/\sqrt{a}$. Their row thresholds increase by $\log(b/a)/(2 \log 2)$ before integer ceilings. This is a direct statement about the same maps and full forms; it supplies no comparison for an unrelated source. The stronger original NG2 same-source width, where applicable, is $O_h(k + \log(q + 2))$. All individual physical masses stay in the constants.

The strengthened results are therefore the complete original-row certificate CS15-20 with arbitrary prescribed accuracy, the explicit full-root coefficient perturbation bounds CS21-23, and the finite original-kernel receiver CS24. Their exact

checks concern finite declared auxiliary data. The native original period-dependent determinant sequence remains unevaluated.

Part IV

The four-label arithmetic receiver

Original four-label ES receiver: full spectrum and first corrections

Independent derivation, 22 September 2026. This file treats the ES sections of `relative_growth_arrival_20260922/03_Pastedtext.txt` and `angular_strengthenin_g_20260922/01_Pastedtext.txt`. The accompanying primary-word resolvent source was read as context; its growing-dimensional RH observation is not identified with this four-label map. The source attribution remains Levent Alpöge's originating announcement, crediting Akhil Mathew and Claude Fable 5, with Tao's exposition credited separately. The following receiver is the later explicit linear extension, not an asserted Jacobian of that originating nonlinear construction.

The parent retains the complete session transcript. The assigned independent calculation is: derive the complete original ES receiver singular asymptotics, family corrections, and relation to the exact supplied inverse column; preserve the original denominators and maps; prove the relation between inverse rates and total volume. No discovery, packaging, publication, or subagent work is undertaken in this derivation.

ES1. Original roots, chart, signs, and exact inverse

Retain four distinct positive labelled roots $\ell = (p, a, Y, Z)$, the original sum $S = p + a + Y + Z$, and

$$f(x) = \prod_j (x - \ell_j), \quad A = -S^{-1}, \quad H = Af, \quad d_j = Af'(\ell_j), \quad \xi_j^2 d_j = 1. \quad (\text{ES1})$$

The source and target coefficient frames are exactly the four labels and the four displayed receiving coordinates. Their Euclidean receiver has columns

$$o_j = \xi_j (1, -id_j, Ad_j^2 + 2\ell_j d_j, i(7\ell_j^2 d_j - 13d_j^2))^T. \quad (\text{ES2})$$

Thus every independent square-root sign is retained. Changing a sign is right multiplication by its diagonal sign isometry and changes the corresponding label coordinate of every inverse and observed vector.

Let e_2, e_3 be the elementary symmetric sums of the literal roots, and put

$$\begin{aligned} P_2(x) &= x^2 + \frac{19S}{7}x + \frac{e_2}{2} - \frac{10S^2}{7}, \quad \mathfrak{h} = \sum_j \frac{P_2(\ell_j)}{d_j^2}, \\ R_1(x) &= i \left[x^3 + \left(\frac{19S^2}{7} + e_2 \right) x - \frac{10S^3}{7} + \frac{Se_2}{2} - \frac{3e_3}{4} \right], \\ R_2(x) &= \frac{5S^2 - 13Sx}{7}, \quad R_3(x) = \frac{i(S - 4x)}{28}, \quad \mathfrak{r}_\nu = \sum_j R_\nu(\ell_j)/d_j^2. \end{aligned} \quad (\text{ES3})$$

On the exact invertibility locus calculated in ES5, the supplied rational expressions are the actual inverse of ES2:

$$(O^{-1}e_0)_j = \xi_j^{-1} \frac{P_2(\ell_j)/d_j^2}{\mathfrak{h}}, \quad (O^{-1})_{j\nu} = \frac{A\xi_j^{-1}}{d_j^2} \left[R_\nu(\ell_j) - P_2(\ell_j) \frac{\mathfrak{r}_\nu}{\mathfrak{h}} \right], \quad 1 \leq \nu \leq 3. \quad (\text{ES4})$$

Here is a direct derivation, retaining all denominators. For target b , put $z_j = \xi_j(O^{-1}b)_j$ and $t_j = d_j z_j$. There is a unique polynomial $L(x) = l_3 x^3 + l_2 x^2 + l_1 x + l_0$ with $t_j = L(\ell_j)/f'(\ell_j)$. The coefficients of the Lagrange identity, or polynomial division by the monic f , give

$$\sum_j \ell_j^n / f'(\ell_j) = 0, 0, 0, 1, S, S^2 - e_2 \quad (0 \leq n \leq 5).$$

Writing $U_L = \sum_j L(\ell_j)$, the last three receiver equations become exactly

$$l_3 = ib_1, \quad A^2 U_L + 2(l_2 + Sl_3) = b_2, \quad 7[l_1 + Sl_2 + (S^2 - e_2)l_3] - 13AU_L = -ib_3,$$

$$U_L = l_3(S^3 - 3Se_2 + 3e_3) + l_2(S^2 - 2e_2) + Sl_1 + 4l_0.$$

The first equation fixes l_3 ; the coefficient determinant in (l_1, l_0) of the other two equations is $(1/S)(52/S) - (4/S^2)20 = -28/S^2 \neq 0$. Hence their homogeneous solution space has dimension one. Substitution verifies that their complete solution is $L = cP_2 + \sum_{\nu=1}^3 b_\nu R_\nu$. The remaining first equation is $A \sum_j L(\ell_j)/d_j^2 = b_0$, which determines $c = (b_0/A - \sum b_\nu \tau_\nu)/\mathfrak{h}$ and proves ES4. This is an exact connecting map from the supplied auxiliary polynomial column to the original labelled inverse.

For the determinant put $\Delta_\ell = \prod_{i < j} (\ell_j - \ell_i)$. Factoring each column by d_j , applying the same polynomial coefficient elimination, and using the Vandermonde determinant gives

$$\det O = 28\epsilon A \left(\prod_j d_j \right) \mathfrak{h}, \quad \epsilon = A^2 \Delta_\ell \prod_j \xi_j \in \{1, -1\}. \quad (\text{ES5})$$

Indeed $\prod_j f'(\ell_j) = \Delta_\ell^2$ for four roots, so $\epsilon^2 = 1$. Equivalently, before any square-root choices, the determinant of the four unscaled columns in ES2 is $28A^3 \Delta_\ell (\prod d_j) \mathfrak{h}$. Thus among all distinct positive labelled roots the exact exceptional receiver space is $\mathfrak{h} = 0$; its complement is precisely the domain of ES4. Below, nonvanishing is proved directly throughout the large- w regime used by the asymptotics. No unproved global certificate at every smaller integral witness is needed for that calculation.

To verify the determinant coefficient explicitly, interpolate $R_0(\ell_j) = 1/d_j$, writing $R_0 = r_0 + r_1 x + r_2 x^2 + r_3 x^3$. After factoring the d_j and cancelling the row constants $(-i)i = 1$, the other three row polynomials are 1 , $A^2 f' + 2x$, and $7x^2 - 13A f'$. The determinant of their four coefficient rows is

$$\frac{28}{S^2} r_1 + \frac{104}{S} r_2 + \left(64 - \frac{14e_2}{S^2} \right) r_3 = 28A^2 \sum_j \frac{P_2(\ell_j) R_0(\ell_j)}{f'(\ell_j)} = 28A^3 \mathfrak{h}.$$

The equality uses exactly the Lagrange sums above. Multiplication by the evaluation determinant Δ_ℓ , by $\prod d_j$, and finally by $\prod \xi_j$ proves ES5 with its stated sign.

The exceptional space also has exact connecting data. At $\mathfrak{h} = 0$, the last three receiver rows still have rank three by the coefficient determinant above. Hence O has rank exactly three, with kernel generated by the nonzero vector $(\xi_j^{-1} P_2(\ell_j)/d_j^2)_j$. Its image is precisely the hyperplane

$$\{b \in \mathbb{C}^4 : b_0/A - \mathfrak{r}_1 b_1 - \mathfrak{r}_2 b_2 - \mathfrak{r}_3 b_3 = 0\}.$$

The defining cokernel row is nonzero because $A \neq 0$. This follows from the first-row compatibility equation after solving all last-three-row equations. Thus the exact defect determines both kernel and cokernel and is not merely a failed inverse formula.

ES2. Uniform large- w exterior spectrum

In the original exterior chart retain

$$R = 4a - p > 0, \quad w = p/R, \quad t = a/p = (1 + w^{-1})/4, \quad u \geq 1, \\ Y = pw(t + t^2/u), \quad Z = pw(t + u), \quad \frac{Y}{Z} = \frac{t}{u}, \quad \frac{YZ}{Y + Z} = pwt. \quad (\text{ES6})$$

These equalities are literal algebraic identities; integrality remains an additional property of the original witness, not of an arbitrary real chart point. For $w \geq 256$, they imply $a < p < Y < Z$, $Y \geq pw/4$, $Z \geq pw$. Set $U = Y + Z$, $\tau = (Z - Y)/U$. Since $Y/Z = t/u \leq 257/1024$, one has $1/2 < \tau < 1$.

Choose $\xi_p = i/\sqrt{-d_p}$, $\xi_a = 1/\sqrt{d_a}$, $\xi_Y = 1/\sqrt{d_Y}$, $\xi_Z = i/\sqrt{-d_Z}$. The exact derivative equations include

$$-d_p = p^2 wt(1 - t) \frac{(1 - p/Y)(1 - p/Z)}{1 + (p + a)/U}, \quad d_a = p^2 wt(1 - t) \frac{(1 - a/Y)(1 - a/Z)}{1 + (p + a)/U}, \\ d_Y = Y^2 \tau \frac{(1 - p/Y)(1 - a/Y)}{1 + (p + a)/U}, \quad -d_Z = Z^2 \tau \frac{(1 - p/Z)(1 - a/Z)}{1 + (p + a)/U}. \quad (\text{ES7})$$

The bounds $p/Y \leq 4/w$, $p/Z \leq 1/w$, $(p + a)/U \leq 4/(3w)$ prove, with constants independent of $u \geq 1$,

$$-d_p, d_a = \frac{3}{16} p^2 w [1 + O(w^{-1})], \quad d_Y = Y^2 \tau [1 + O(w^{-1})], \quad -d_Z = Z^2 \tau [1 + O(w^{-1})]. \quad (\text{ES8})$$

All estimates in this section are uniform for $p \geq 1, u \geq 1, w \geq 256$; no nearly coincident-tail denominator is hidden in them.

For completeness $\mathfrak{h} < 0$ on this entire regime. Positivity of the four roots gives $e_2 \leq 3S^2/8$, while $p/S \leq 1/w$. Consequently both core values satisfy $P_2(p), P_2(a) \leq -S^2$. ES7 gives $|d_p|, d_a \leq p^2 w/4$, so the core contribution to $\mathfrak{h}$ is at most $-32S^2/(p^4 w^2)$. At every root $|P_2(\ell_j)| \leq 6S^2$. ES7 gives $d_Y \geq Y^2/3$, $-d_Z \geq Z^2/3$, hence the absolute tail contribution is at most

$$54S^2(Y^{-4} + Z^{-4}) \leq 27648S^2/(p^4 w^4).$$

The ratio of this last bound to the core magnitude is at most $864/w^2 < 1$. This proves nonvanishing of ES4–5 directly, without relying on an unexecuted global coefficient certificate.

Define the positive literal-tail quantities

$$A_3 = \sqrt{\tau[Y^6(7 - 13\tau)^2 + Z^6(7 + 13\tau)^2]}, \\ A_{23} = \tau^2 Y^2 Z^2 U \left(40 - \frac{14YZ}{U^2} \right). \quad (\text{ES9})$$

The bracket is at least $73/2$. The complete increasingly numbered by decreasing size singular values are

$$s_1 = A_3[1 + O(w^{-1})], \quad s_2 = (A_{23}/A_3)[1 + O(w^{-1})], \\ s_3 = \sqrt{3/8} p \sqrt{w} [1 + O(w^{-1})], \quad s_4 = \sqrt{32/3} (p \sqrt{w})^{-1} [1 + O(w^{-1})]. \quad (\text{ES10})$$

Here and below singular values use the original Euclidean label and receiving metrics, until the explicit transport ES33–37 in ES_FAMILY_CORRECTIONS.md.

We prove all four rates by the complete exterior maps. The leading tail block in rows 2 and 3 is

$$\begin{pmatrix} Y^2\sqrt{\tau}(2 - Y\tau/U) & -iZ^2\sqrt{\tau}(2 + Z\tau/U) \\ iY^3\sqrt{\tau}(7 - 13\tau) & Z^3\sqrt{\tau}(7 + 13\tau) \end{pmatrix}. \quad (\text{ES11})$$

Its determinant is exactly A_{23} : expanding the two terms reduces the remaining bracket to $\tau U(40 - 14YZ/U^2)$. ES8 gives errors bounded entrywise by $CY^2/w, CZ^2/w, CY^3/w, CZ^3/w$, respectively. The last-row norm is $A_3[1 + O(w^{-1})]$; its Z term is uniformly nonzero even if $7 - 13\tau$ vanishes.

Here are the bounds for every omitted exterior coefficient. Core entries in rows 0,1,2,3 have respective sizes at most $C/(p\sqrt{w}), Cp\sqrt{w}, Cp^2\sqrt{w}, Cp^3w^{3/2}$; tail entries in those rows have respective sizes at most $C/L, CL, CL^2, CL^3$, for $L = Y, Z$. Relative to $A_{23} \geq (73/8)Y^2Z^3$, any other two-column minor is bounded by C/w : for tail columns lowering row 2 costs at most C/Y , and for one core column the largest ratio is bounded by $C[p^2\sqrt{w}/Y^2 + p^3w^{3/2}/(Y^2Z)]$. Both are at most C/w ; two core columns are smaller. The finite number of minors then bounds the entire second exterior map, not only one chosen coefficient.

At exterior rank three the two leading entries, in rows 1,2,3 and columns (p, Y, Z) , (a, Y, Z) , are

$$-\frac{\sqrt{3}}{4}p\sqrt{w}A_{23}[1 + O(w^{-1})], \quad -\frac{i\sqrt{3}}{4}p\sqrt{w}A_{23}[1 + O(w^{-1})].$$

The other terms in either same minor cost at most $Cp/Y + Cp^2w/(YZ) = O(w^{-1})$ relatively. A triple with two core columns has relative size at most $Cw^{-3/2}$, and replacing row 1 by row 0 costs at most $C/(p^2w)$. Thus the leading exterior map has one nonzero row with both displayed entries, giving

$$\|\wedge^3 O\| = \sqrt{3/8}p\sqrt{w}A_{23}[1 + O(w^{-1})]. \quad (\text{ES12})$$

For the full determinant the core rows 0,1 minor equals $(d_a - d_p)/\sqrt{-d_p d_a} = 2 + O(w^{-1})$. The largest alternative pairing, core rows 0,2 and tail rows 1,3, has relative size at most Cp/Y ; core rows 0,3 and tail rows 1,2 cost at most $Cp^2w/(YZ)$. The remaining pairings cost at most these bounds. Therefore

$$|\det O| = 2A_{23}[1 + O(w^{-1})]. \quad (\text{ES13})$$

The identities $\|\wedge^j O\| = s_1 \cdots s_j$ now prove ES10 with all four singular values retained.

The same calculation, or ES4 with ES7, gives the inverse operator limit

$$\frac{O^{-1}}{p\sqrt{w}} = \frac{\sqrt{3}}{8}(-i, 1, 0, 0)^\top e_0^* + O(w^{-1}) \quad (\text{ES14})$$

in operator norm, uniformly in the stated regime. To see why all columns are controlled, the cofactor of column zero has the two ES12 leading triples, while every cofactor of column one has norm at most $CA_{23}/(p\sqrt{w})$; columns two and three have at most $CA_{23}/(p^2\sqrt{w})$ and $CA_{23}/(p^3w^{3/2})$, respectively. Division by ES13 and then by $p\sqrt{w}$ makes all these columns $O(w^{-1})$, uniformly for $p \geq 1$. The leading two zero-column cofactors have the phases in ES14; their exact formula is ES4.

Let $v_0 = 2^{-1/2}(-i, 1, 0, 0)^\top$. Rank-one singular perturbation of ES14 and ES10 prove, for every fixed $\tau_0 > 0$,

$$e^{-\tau_0 p^2 w O^* O} \longrightarrow e^{-(32/3)\tau_0} v_0 v_0^*, \quad (\text{ES15})$$

uniformly as $w \rightarrow \infty$ in this original chart. The other three exponents diverge because $p^2 w s_3^2 \asymp p^4 w^2 \rightarrow \infty$. The four marked labels remain distinct in the limiting projection.

Original ES family: complete first corrections and observation hierarchy

This is the continuation ES16-37 of ES_PROOFS.md, 22 September 2026. Every root, map, branch, and denominator is ES1-7 from that proof.

ES3. Exact fixed- c integral family and its progression

Fix an integer $c \geq 2$, $c \equiv 2 \pmod{3}$. At primes satisfying $p \equiv 1 \pmod{12}$, $c \mid a = (p+3)/4$, retain

$$T = pa, \quad Y = (T + c)/3, \quad Z = T(T + c)/(3c). \quad (\text{ES16})$$

Since $p \equiv a \equiv 1 \pmod{3}$, $T + c \equiv 0 \pmod{3}$; since $c \mid T$, both Y, Z are integers. Direct fractions give $Y^{-1} + Z^{-1} = 3/T$ and $a^{-1} + 3/(pa) = 4/p$. Thus this is an existing integral ES witness in the original chart, with $R = 3, w = p/3, u = a^2/c \geq 1$ eventually. No occupancy assertion for every prime in a residue class is made.

The explicit choice $c = 29$, $p \equiv 3361 \pmod{24360}$ satisfies $p \equiv 1 \pmod{840}$, $p \equiv -3 \pmod{116}$, and $\gcd(3361, 24360) = 1$. The classical coprime-progression theorem of P. G. Lejeune-Dirichlet gives infinitely many primes in this progression. The retained source is the 1837 paper in Ralf Stephan's English translation, arXiv:0808.1408v2; its opening theorem and translator credit were read directly in the retained TeX. This translation is not described as Dirichlet's original German typesetting. The initial prime 3361 gives exactly

$$(p; a, Y, Z) = (3361; 841, 942210, 91836266490), \quad T = 2826601. \quad (\text{ES17})$$

ES4. Full analytic matrix and every first singular correction

Use $z = 1/p$ here, to distinguish the expansion variable from the chart's $t = a/p$. Define the exact weighted matrix

$$B(z) = \text{diag}(1, p^{-3}, p^{-4}, p^{-6}) O_p \text{diag}(p^{3/2}, p^{3/2}, 1, p^{-6}). \quad (\text{ES18})$$

The roots and d_j are rational functions of z . Their four derivative leading terms are $-p^3/16, p^3/16, p^4/144, -p^8/(2304c^2)$. Factoring these terms out of the prescribed square roots leaves analytic functions equal to one at zero. Every exponent in ES18 is then integral and nonnegative. Hence B is analytic at zero, with a convergent local Taylor series for each fixed c .

Direct substitution gives its complete first two coefficients:

$$B_0 = \begin{pmatrix} 4i & 4 & 0 & 0 \\ -1/4 & -i/4 & 0 & 0 \\ -i/2 & 1/8 & 1/72 & 0 \\ 13/64 & -13i/64 & -i/288 & 5/(27648c^3) \end{pmatrix},$$

$$B_1 = \begin{pmatrix} 20i & 2 & 0 & 0 \\ 5/4 & i/8 & -i/12 & 0 \\ 5i/2 & 5/16 & -1/48 & 0 \\ -83/64 & 53i/128 & 31i/288 & 5/(1536c^3) \end{pmatrix}, \quad B(z) = B_0 + zB_1 + O_c(z^2). \quad (\text{ES19})$$

In particular $\det B_0 = 5/(995328c^3) > 0$. Its inverse is analytic, and

$$B(z)^{-1}e_0 = z_0 + zz_1 + O_c(z^2),$$

$$z_0 = (-i/8, 1/8, 27/8, 1728ic^3/5)^\top, \quad z_1 = (-i/2, -19/16, 135/16, -50112ic^3/5)^\top. \quad (\text{ES20})$$

These are verified by $B_0 z_0 = e_0$, $B_0 z_1 = -B_1 z_0$, giving their exact derivation without an inferred inverse rate. In particular the nonzero entry $(B_1)_{1Y} = -i/12$ and the original Y-coordinate $27/8$ both enter the two core derivatives.

The exact inverse factorization is

$$O_p^{-1} = \text{diag}(p^{3/2}, p^{3/2}, 1, p^{-6}) B (1/p)^{-1} \text{diag}(1, p^{-3}, p^{-4}, p^{-6}). \quad (\text{ES21})$$

Thus its first column has all four expansions

$$\begin{aligned} (O_p^{-1} e_0)_p &= -\frac{i}{8} p^{3/2} - \frac{i}{2} p^{1/2} + O_c(p^{-1/2}), & (O_p^{-1} e_0)_a &= \frac{1}{8} p^{3/2} - \frac{19}{16} p^{1/2} + O_c(p^{-1/2}), \\ (O_p^{-1} e_0)_Y &= \frac{27}{8} \left(1 + \frac{5}{2p} + O_c(p^{-2}) \right), & (O_p^{-1} e_0)_Z &= \frac{1728ic^3}{5} p^{-6} \left(1 - \frac{29}{p} + O_c(p^{-2}) \right). \end{aligned} \quad (\text{ES22})$$

These expand the exact ES4 column, with the same roots, derivative denominators and square-root choices. They are not a different auxiliary receiver.

Every singular value has its first correction evaluated:

$$\begin{aligned} s_1(O_p) &= \frac{5}{27648c^3} p^{12} \left(1 + \frac{18}{p} + O_c(p^{-2}) \right), \\ s_2(O_p) &= \frac{1}{72} p^4 \left(1 - \frac{3}{2p} + O_c(p^{-2}) \right), \\ s_3(O_p) &= \frac{1}{2\sqrt{2}} p^{3/2} \left(1 - \frac{41}{4p} + O_c(p^{-2}) \right), \\ s_4(O_p) &= 4\sqrt{2} p^{-3/2} \left(1 + \frac{11}{4p} + O_c(p^{-2}) \right). \end{aligned} \quad (\text{ES23})$$

The complete determinant, with the retained branches, is

$$\det O_p = \frac{5}{995328c^3} p^{16} \left(1 + \frac{9}{p} + O_c(p^{-2}) \right). \quad (\text{ES24})$$

We prove the errors as well as the coefficients. The leading entry of O_p is row 3, column Z, namely $p^{12} B_{3Z}$; every other entry is at most $O_c(p^8)$. Thus its norm has the relative first correction 18 by ES19. For exterior rank two, the leading minor is rows 2,3 and columns Y,Z:

$$p^{16} \det B_{\{2,3\},\{Y,Z\}} = \frac{5}{1990656c^3} p^{16} \left(1 + \frac{33}{2p} + O_c(p^{-2}) \right).$$

Every other minor is at most $O_c(p^{15})$. Its squared Frobenius norm is therefore the squared leading minor times $1 + O_c(p^{-2})$; the operator norm lies between that leading modulus and the Frobenius norm. This proves the stated first correction for $s_1 s_2$.

For exterior rank three, both leading minors are in rows 1,2,3 and columns (p, Y, Z) , (a, Y, Z) , of scale $p^{35/2}$. Their exact analytic coefficients are the corresponding minors $M_p(z)$, $M_a(z)$ of B . Expanding ES19 gives

$$\frac{\frac{d}{dz} (|M_p(z)|^2 + |M_a(z)|^2)}{2(|M_p(0)|^2 + |M_a(0)|^2)} \Big|_{z=0} = \frac{25}{4}.$$

The leading norm is $5p^{35/2}/(3981312\sqrt{2}c^3)$. Every other entry is at most $O_c(p^{16})$, so its squared contribution is relatively $O_c(p^{-3})$. The norm of the complete leading row is a lower bound, and the Frobenius norm supplies the matching upper bound. Thus

the first correction to $s_1 s_2 s_3$ is $25/4$. Finally $\det O_p = p^{16} \det B(1/p)$, and $\text{Tr}(B_0^{-1} B_1) = 9$. Taking successive exterior ratios gives $18, 33/2 - 18 = -3/2, 25/4 - 33/2 = -41/4, 9 - 25/4 = 11/4$, proving ES23-24.

A second route to the smallest singular value uses $u_0 = (-i, 1, 0, 0)^T/8$, $u_1 = (-i/2, -19/16, 0, 0)^T$. ES21 gives the first normalized inverse column $u_0 + p^{-1}u_1 + O_c(p^{-3/2})$; its $p^{-3/2}$ term is in the orthogonal Y coordinate. The other normalized inverse columns are $O_c(p^{-3})$. Since $\|u_0\|^2 = 1/32$ and $2\Re(u_0^* u_1) = -11/64$,

$$p^{-3}\|O_p^{-1}\|^2 = \frac{1}{32} - \frac{11}{64p} + O_c(p^{-2}), \quad \boxed{p^3 s_4(O_p)^2 = 32 + \frac{176}{p} + O_c(p^{-2})}. \quad (\text{ES25})$$

ES5. Exact slow-vector hierarchy and scalar observations

Let v_p be the right singular vector of O_p for s_4 , with its a coordinate positive. Put $v_0 = (-i, 1, 0, 0)^T/\sqrt{2}$, $w_0 = (i, 1, 0, 0)^T/\sqrt{2}$. Then

$$v_p = v_0 - \frac{27}{4p}w_0 + \frac{27\sqrt{2}}{2p^{3/2}}e_Y + O_c(p^{-2}). \quad (\text{ES26})$$

Both tail coordinates have stronger relative expansions:

$$\boxed{e_Y^* v_p = \frac{27\sqrt{2}}{2} p^{-3/2} \left(1 + \frac{21}{4p} + O_c(p^{-2}) \right)},$$

$$\boxed{e_Z^* v_p = \frac{6912\sqrt{2}ic^3}{5} p^{-15/2} \left(1 - \frac{105}{4p} + O_c(p^{-2}) \right)}. \quad (\text{ES27})$$

To prove relative accuracy, write exactly

$$p^{-3/2}O_p^{-1} = D(z)B(z)^{-1}E(z), \quad D(z) = \text{diag}(1, 1, z^{3/2}, z^{15/2}), \quad E(z) = \text{diag}(1, z^3, z^4, z^6).$$

The right Gram of this inverse has a simple leading eigenvalue $1/32$. Its top eigenvector has coordinates $(1, O_c(z^3), O_c(z^4), O_c(z^6))$, up to scalar normalization: solve the three lower block equations, whose scalar eigenvalue stays separated from zero. Multiplication by E changes the coefficient vector from e_0 by $O_c(z^6)$. The left singular vector before normalization is therefore $D(z)[B(z)^{-1}e_0 + O_c(z^6)]$, preserving the powers in each tail coordinate. Its norm is $\|u_0\|[1 - 11z/4 + O_c(z^2)]$. ES20-22 give ES26 and the relative coefficients $5/2 + 11/4 = 21/4$, $-29 + 11/4 = -105/4$ in ES27. A tiny component is not inferred from a larger vector error.

For $\mathcal{H}_p(\tau) = e^{-\tau p^3 O_p^* O_p}$, the three faster directions are $O_c(e^{-c_1 \tau p^6})$, while ES25 supplies the slow exponential. For each fixed nonzero scalar row ℓ , the following tests are exhaustive:

Exact original-label test	Power ν	Limit of $p^\nu \ell \mathcal{H}_p(\tau) \ell^*$
$\ell v_0 \neq 0$	0	$ \ell v_0 ^2 e^{-32\tau}$
$\ell v_0 = 0, \ell w_0 \neq 0$	2	$(729/16) \ell w_0 ^2 e^{-32\tau}$
$\ell v_0 = \ell w_0 = 0, \ell e_Y \neq 0$	3	$(729/2) \ell e_Y ^2 e^{-32\tau}$
$\ell v_0 = \ell w_0 = \ell e_Y = 0$	15	$(95551488c^6/25) \ell e_Z ^2 e^{-32\tau}$

The vectors v_0, w_0 span exactly the original core coordinates, so no nonzero row is missing. Convergence is locally uniform in $\tau \in (0, \infty)$, without an assertion at zero time. The two pure tail observations also have evaluated first corrections:

$$p^3 e_Y^* \mathcal{H}_p(\tau) e_Y = \frac{729}{2} e^{-32\tau} \left[1 + \frac{21/2 - 176\tau}{p} + O_c(p^{-2}) \right],$$

$$p^{15}e_Z^*\mathcal{H}_p(\tau)e_Z = \frac{95551488c^6}{25}e^{-32\tau}\left[1 - \frac{105/2 + 176\tau}{p} + O_c(p^{-2})\right]. \quad (\text{ES28})$$

These follow by squaring ES27 and multiplying the exponential from ES25; the faster modes remain below the remainder after either polynomial rescaling.

For the actual two-row observation $L_{\text{obs}} = \begin{pmatrix} 1/\sqrt{2} & i/\sqrt{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$, $D_p = \text{diag}(p, p^{3/2})$, its first row annihilates v_0 and takes value i on w_0 . Hence

$$D_p L_{\text{obs}} \mathcal{H}_p(\tau) L_{\text{obs}}^* D_p \longrightarrow e^{-32\tau} \begin{pmatrix} 729/16 & -729i\sqrt{2}/8 \\ 729i\sqrt{2}/8 & 729/2 \end{pmatrix}. \quad (\text{ES29})$$

Replacing the core phases by their moduli before applying this observation would destroy its cancellation.

ES6. Same inverse limits, different complete volume on the same map

For primes $p \equiv 13 \pmod{24}$, retain the exact original-chart family

$$a = (p+3)/4, \quad Y = (p+3)(3p+1)/32, \quad Z = p(3p+1)/4. \quad (\text{ES30})$$

Writing $p = 24j + 13$ gives $p+3 = 8(3j+2)$, $3p+1 = 8(9j+5)$, proving integrality. Direct substitution gives $Y^{-1} + Z^{-1} = 12/[p(p+3)] = 3/(pa)$. Thus the same ES identity and receiver ES2 apply, with $R = 3, w = p/3, u = 2$; the initial witness is $(13; 4, 20, 130)$.

Here $Y/p^2 \rightarrow 3/32$, $Z/p^2 \rightarrow 3/4$, $\tau \rightarrow 7/9$. Substituting into the literal ES9 quantities gives

$$s_1 = \frac{7\sqrt{55508999}}{8192}p^6[1 + O(p^{-1})], \quad s_2 = \frac{2737}{24\sqrt{55508999}}p^4[1 + O(p^{-1})],$$

$$s_3 = \frac{1}{2\sqrt{2}}p^{3/2}[1 + O(p^{-1})], \quad s_4 = 4\sqrt{2}p^{-3/2}[1 + O(p^{-1})], \quad |\det O_p| = \frac{19159}{98304}p^{10}[1 + O(p^{-1})]. \quad (\text{ES31})$$

Both families have the same limits

$$p^{-3/2}\|O_p^{-1}\| \rightarrow 1/(4\sqrt{2}), \quad \|\wedge^2 O_p^{-1}\| \rightarrow 1/2, \quad e^{-\tau p^3 O_p^* O_p} \rightarrow e^{-32\tau} v_0 v_0^*. \quad (\text{ES32})$$

Their complete determinants have different powers, p^{10} and p^{16} . This compares existing integral families under the same receiver; it does not invent an unrelated spectrum. Their exact relation is $|\det O| = (s_1 s_2) / \|\wedge^2 O^{-1}\|$: the two inverse limits control s_3, s_4 , while the original tail factor $A_{23} = s_1 s_2 [1 + O(w^{-1})]$ supplies the missing volume.

ES7. Exact fixed-metric and conductor transport

For fixed positive label metric G and fixed positive receiving metric Q , the original energy is $H_{G,Q} = G^{-1}O_p^* Q O_p$. Its orthonormal representative is $\widehat{O} = Q^{1/2}O_p G^{-1/2}$, with exact inverse $G^{1/2}O_p^{-1}Q^{-1/2}$. Put $n_G = u_0^* G u_0 > 0$, $q_Q = e_0^* Q^{-1} e_0 > 0$. ES21 gives the rank-one inverse limit $G^{1/2}u_0 e_0^* Q^{-1/2}$, whence

$$p^3 \lambda_{\min}(H_{G,Q}) \longrightarrow (n_G q_Q)^{-1}. \quad (\text{ES33})$$

Its metric-dependent first correction is

$$p^3 \lambda_{\min}(H_{G,Q}) = \frac{1}{n_G q_Q} \left[1 - \frac{2\Re(u_0^* G u_1)}{n_G p} + O_{c,G,Q}(p^{-3/2}) \right]. \quad (\text{ES34})$$

Indeed the first inverse-column squared G norm is $n_G + 2\Re(u_0^* G u_1)/p + O(p^{-3/2})$. Other inverse columns enter the weighted target contraction first at order p^{-3} . The simple leading eigenvalue expansion, obtained from its complementary block Schur equation, yields ES34. A $p^{-3/2}$ term may occur when G has nonzero core-Y entries; the stronger Euclidean remainder is not assigned to arbitrary metrics.

The observed covariance $\ell e^{-\tau p^3 H_{G,Q}} G^{-1} \ell^*$ has the same four powers 0, 2, 3, 15, tested successively by ℓu_0 , ℓu_1 , ℓe_Y , ℓe_Z . Its respective leading numerators are

$$|\ell u_0|^2, \quad |\ell u_1|^2, \quad |\ell(27e_Y/8)|^2, \quad |\ell(1728ic^3e_Z/5)|^2, \quad (\text{ES35})$$

each divided by n_G and multiplied by $e^{-\tau/(n_G q_Q)}$, after the preceding numerators vanish. For relative last-coordinate accuracy, write the inverse exactly as $G^{1/2} D(z) B(z)^{-1} E(z) Q^{-1/2}$. Its top left singular vector, returned to label coordinates, is $D(z)[B(z)^{-1} e_0 + O_{G,Q}(z^3)]$, up to normalization: the off-diagonal terms in $E Q^{-1} E$ first occur at order z^3 . The Y and Z coordinate powers therefore stay $z^{3/2}$, $z^{15/2}$, with the same nonzero leading coefficients. Normalization terms multiply earlier vectors and cancel under the successive annihilation tests. This proves the weighted hierarchy without imposing Euclidean orthogonality after transport.

The attained scalar observation metric is $Q_\ell = (\ell G^{-1} \ell^*)^{-1}$; multiply the covariance by it to obtain measured scalar heat. On a fixed admitted invertible conductor restriction L , take the actual metric $Q = L^* Q_\Gamma L$. Then $q_Q = e_0^* (L^* Q_\Gamma L)^{-1} e_0$ retains the full conductor inverse, Gamma mass, and period data.

For moving metrics or conductor restrictions, the underlying error transport is

$$\begin{aligned} & \|G^{1/2}(O^{-1}/(p\sqrt{w}) - a_* e_0^*) Q^{-1/2}\| \\ & \leq \sqrt{\lambda_{\max}(G)/\lambda_{\min}(Q)} \|O^{-1}/(p\sqrt{w}) - a_* e_0^*\|, \\ & a_* = \sqrt{3}(-i, 1, 0, 0)^T/8. \end{aligned} \quad (\text{ES36})$$

Composition by the actual invertible conductor L has relative leading-operator error bounded by

$$\frac{\|O^{-1}/(p\sqrt{w}) - a_* e_0^*\| \|L^{-1}\|}{\|a_*\| \|e_0^* L^{-1}\|}. \quad (\text{ES37})$$

This denominator is part of the proved connecting map. A conductor approaching a directional resonance need not have a uniform bound here. Neither the growing RH kernel coefficient nor a complex-current phase has been assigned from the ES soft mode.

Verification

The exact checker derives ES19–20, all exterior first corrections, the complete rational inverse, integral families, and initial witness. Independent 110-digit SVD checks retain both core triples and test the four first singular corrections, soft energy, relative last-label component, and distinct-volume family. Large integer samples used for analytic asymptotics are not asserted to be primes. Source versions, reading coverage, and original/translated source distinctions are recorded in the accompanying ledger.

Part V

The received primary-word determinant calculation

The original primary-word measured resolvent

Complete additive proofs and original-source extraction

21 September 2026. Equations PR1-PR44, including lettered extensions.

The finite identities below are new derivations on the specified maps. Arithmetic asymptotics consume the two retained uploads and the pinned DCR source with their stated domains and analytic-input status. All roots, source masses, the original observation, full minima, and zero directions remain. The original kernel coefficient is not assigned. See README.md, state/READING_LEDGER.json, and state/VERIFICATION.json.

PR1-PR16. The full measured primary-word resolvent

21 September 2026. New finite derivation. These identities use the specified positive metric, linear word, and onto observation. They do not use a monic-profile asymptotic. Their arithmetic specialization uses the two retained input files. Adjoints are metric adjoints; ordinary stars are conjugate transposes in the coefficient frames that are explicitly displayed.

PR1. Original maps and unchanged values

Let E have dimension q and positive metric G . Retain the actual onto observation $\Lambda: E \rightarrow B$, of rank n , its kernel K of dimension $m=q-n$, and the fixed kernel frame I_K . Define

$$Q_B = (\Lambda G^{-1} \Lambda^*)^{-1}, \quad L = G^{-1} \Lambda^* Q_B, \quad H_K = I_K^* G I_K. \quad (\text{PR1})$$

Then $\Lambda L = I$, $L^* G L = Q_B$, and $L \Lambda$ is the G -orthogonal projection onto $K^\perp$. The isometries from Euclidean coefficient spaces are

$$J_B = L Q_B^{-1/2}, \quad J_K = I_K H_K^{-1/2}.$$

For the retained linear word T , put $H = T^\dagger T$, $V = \ker T$, $d = \dim V$, $\delta = q - d$. The observed endomorphism to be calculated is

$$\mathcal{Y}(z) = \Lambda z (zI + H)^{-1} L, \quad z > 0. \quad (\text{PR2})$$

It is positive and at most I in Q_B . Its Euclidean representation is $Q_B^{(1/2)} \text{math-scrY} Q_B^{(-1/2)} = J_B^\dagger z(zI + H)^{-1} J_B$. The determinant is unchanged by that similarity. Moreover

$$\mathcal{Y}(z) = \int_0^\infty z e^{-z\tau} \Lambda e^{-\tau H} L d\tau. \quad (\text{PR3})$$

The integral is the scalar identity on the complete positive spectral resolution of H , including its d zero eigenvalues. No projection is moved through H .

PR2. The exact overlap and all zero factors

Choose any isometry $J_{V^\perp}$ onto $V^\perp$, and define

$$B_+ = J_{V^\perp}^\dagger H J_{V^\perp} > 0, \quad C = J_K^\dagger J_{V^\perp}, \quad A_K = J_K^\dagger H J_K = C B_+ C^*, \quad \Gamma = C C^*. \quad (\text{PR4})$$

Write $R = J_{V^\perp}^\dagger J_{V^\perp}$. Because $J_{V^\perp}^\dagger J_{V^\perp} = I$, $RR = I - CC$. Direct substitution in PR2 gives the complete original observed matrix

$$\widehat{\mathcal{Y}}(z) = I - R B_+ (zI + B_+)^{-1} R^*. \quad (\text{PR5})$$

The factors C and B_+ need not commute. Sylvester's determinant identity, proved by the determinant of the block matrix $\begin{bmatrix} I & U \\ -V & I \end{bmatrix}$, gives

$$\det_B \mathcal{Y}(z) = z^{\delta-m} \frac{\det(zI_m + A_K)}{\det(zI_\delta + B_+)}. \quad (\text{PR6})$$

Indeed its delta-dimensional numerator is $zI + B_+^{(1/2)} C C^* B_+^{(1/2)}$, whose nonzero Gram spectrum equals that of $C B_+ C$. In the original fixed coefficient frame the identical formula is

$$\det_B \mathcal{Y}(z) = z^{q-m} \frac{\det(zH_K + I_K^* T^* G T I_K)}{\det H_K \det(zI_q + G^{-1} T^* G T)}. \quad (\text{PR7})$$

PR7 uses only matrices invertible for $z > 0$ and has no positive-range convention. Let $h = \dim(K \cap V)$, $p = m - h$. Then $\text{rank } A_K = p$, and PR6 is equivalently

$$\det_B \mathcal{Y}(z) = z^{\delta-p} \frac{\prod_{j=1}^p (z + \alpha_j)}{\prod_{j=1}^\delta (z + \lambda_j)}, \quad (\text{PR8})$$

where α_j are all positive eigenvalues of A_K and λ_j all positive eigenvalues of H . The forced power is $\delta - m + h$, not $\delta - m$ with h dropped. $\text{Rank}(T|K) = m - h \leq \delta$ proves that this exponent is nonnegative.

When parameters change rank, PR7 remains the single exact formula. A newly zero numerator eigenvalue contributes a factor z ; a newly zero denominator eigenvalue also contributes its own z . No disappearing direction is assigned value one.

Exact complementary factor at zero regularizer

On the original transverse domain $h=0$ let $U = C \Gamma^{(-1/2)}$; it is an isometry onto $\text{ran}(C)$. Choose an isometry Z onto $\ker C$, of dimension $\rho = \delta - m$. Full block inversion in $\begin{bmatrix} U & Z \end{bmatrix}$ gives

$$\frac{\det A_K}{\det B_+} = \det \Gamma \det \Xi, \quad \Xi = Z^* B_+^{-1} Z. \quad (\text{PR8a})$$

In fact $\det A_K = \det \Gamma \det(U B_+ U)$, and the lower inverse block is the inverse of the COMPLETE Schur complement of $U B_+ U$. This proves the ratio without replacing the off-diagonal blocks by zero. For an arbitrary full frame Z_0 of $\ker C$, the identical last determinant is $\det(Z_0 B_+^{-1} Z_0) / \det(Z_0 Z_0)$. Every inverse and every Gram in this comparison is specified.

PR3. Positive spectral shift and every regularizer derivative

Because $\|C\| \leq 1$, singular-value min-max gives $\alpha_j \leq \lambda_j$ when both positive lists are decreasing and $j \leq p$. Thus

$$\xi(s) = \#\{j : \lambda_j > s\} - \#\{j : \alpha_j > s\}$$

satisfies $0 \leq \xi(s) \leq \delta$. Scalar integration in PR8 yields

$$-\log \det_B \mathcal{Y}(z) = \int_0^\infty \frac{\xi(s)}{z+s} ds. \quad (\text{PR9})$$

This identity has every original eigenvalue, including multiplicity. It proves, for every integer $j \geq 1$,

$$0 \leq (-1)^j \frac{d^j}{dz^j} [-\log \det \mathcal{Y}(z)] \leq \frac{\delta(j-1)!}{z^j}. \quad (\text{PR10})$$

In particular $0 \leq z(d/dz) \log \det \mathcal{Y}(z) \leq \delta$. These are uniform in the positive metric and in the nonzero spectral gaps. With $a = \min \lambda_j$ and $b = \max \lambda_j$, $\xi(s) \geq \delta - p$ on $0 < s < a$, so

$$-\delta \log(1 + b/z) \leq \log \det \mathcal{Y}(z) \leq -(\delta - p) \log(1 + a/z). \quad (\text{PR11})$$

The exact large- z expansion retains the differences of full moments:

$$-\log \det \mathcal{Y}(z) = \sum_{j=1}^N \frac{(-1)^{j+1}}{j z^j} [\text{Tr } B_+^j - \text{Tr } A_K^j] + \mathcal{E}_N(z).$$

The remainder has the sign $(-1)^N$ and obeys

$$|\mathcal{E}_N(z)| \leq \frac{\text{Tr } B_+^{N+1} - \text{Tr } A_K^{N+1}}{(N+1)z^{N+1}}. \quad (\text{PR12})$$

This follows by finite division of $1/(z+s)$, then integration against $\xi(s)$; it is valid even when its numerical bound is large. In the first moment the exact difference is $\text{Tr}(P_B H)$. At higher powers $\text{Tr}(A_K^j)$ is not replaced by $\text{Tr}(P_K H^j)$.

The matrix-valued regularizer derivatives retain every mixed block

Let P_λ be the full positive spectral projections of H and put $B_\lambda = J_B^\dagger P_\lambda J_B \succeq 0$. In the exact observed orthonormal frame,

$$\widehat{\mathcal{Y}}(z) = Y_0 + \sum_{\lambda>0} \frac{z}{z+\lambda} B_\lambda, \quad \sum_{\lambda>0} B_\lambda = I - Y_0. \quad (\text{PR12a})$$

The matrices B_λ need not commute. All derivatives, with these matrices unchanged, satisfy

$$0 \preceq z^j (-1)^{j-1} \widehat{\mathcal{Y}}^{(j)}(z) \preceq j! \frac{z^j}{(j+1)^{j+1}} (I - Y_0), \quad j \geq 1. \quad (\text{PR12b})$$

This is the maximum of $x/(1+x)^{(j+1)}$, at $x=1/j$. For $\text{Im } z > 0$,

$$\text{Im } \widehat{\mathcal{Y}}(z) = (\text{Im } z) \sum_{\lambda>0} \frac{\lambda}{|z+\lambda|^2} B_\lambda \succeq 0. \quad (\text{PR12c})$$

It is invertible there: a kernel vector would annihilate every B_λ by the imaginary quadratic form; then Y_0 acts as the identity on it, a contradiction. Thus this is a complete holomorphic matrix family, with the displayed mixed coefficients, on the upper half-plane.

PR4. Scalar spectral enclosure with the complete overlap retained

Retain any finite c and $E \geq 0$ for which

$$e^{c-E}I \preceq B_+ \preceq e^{c+E}I.$$

This is a concrete finite input in the original application. Let $Y_0 = J_B^\dagger P_V J_B$. Then

$$Y_0 + \frac{z}{z + e^{c+E}}(I - Y_0) \preceq \widehat{\mathcal{Y}}(z) \preceq Y_0 + \frac{z}{z + e^{c-E}}(I - Y_0). \quad (\text{PR13})$$

The central model is $\bar{Y} = Y_0 + z/(z + \exp(c))(I - Y_0)$. PR13 also yields the relative form comparison $\exp(-E)\bar{Y} \leq \widehat{\text{mathscrY}} \leq \exp(E)\bar{Y}$. Every fixed restriction and every full attained quotient of these two positive forms inherits that same comparison by minimizing on the identical affine fibres. Its rank- p determinant error is at most pE . No new quotient minimum is assigned a diagonal metric.

The common unit subspace $\ker(I - Y_0)$ is fixed pointwise by all three operators. Its complementary dimension is at most δ . For $0 \leq \mu \leq 1$,

$$f_a(\mu) = (z + a\mu)/(z + a)$$

has $|\log f_a / \log a| \leq 1$. Therefore the logarithm of each nonunit eigenvalue of mathscrY differs by at most E from the same ordered eigenvalue of $Y_0 + z/(z + \exp(c))(I - Y_0)$. The determinant error is at most δE , not qE .

Let $\gamma_1, \dots, \gamma_p$ be all positive eigenvalues of Γ . Its h zero eigenvalues stay separate. The exact flat-spectrum determinant is

$$\det[Y_0 + \eta(I - Y_0)] = \eta^{\delta-p} \prod_{j=1}^p [\eta + (1 - \eta)\gamma_j], \quad 0 < \eta \leq 1. \quad (\text{PR14})$$

It follows directly from PR5 with the scalar $B_+ = aI$. It remains valid when some γ_j equal one, or when $\delta < m$. No negative unit multiplicity need be used.

Fix a positive scale Q and put $z = \exp(c - Qt)$, $\kappa_j = -(\log \gamma_j)/Q \geq 0$. The elementary inequality $0 \leq \log(1 + \exp x) - \max(x, 0) \leq \log 2$ gives

$$\left| \log \det \mathcal{Y}(z) + Q \left[(\delta - p)t_+ + \sum_{j=1}^p \min(t_+, \kappa_j) \right] \right| \leq \delta(E + \log 2). \quad (\text{PR15})$$

The bound is uniform for all real t , including moving parameters and thresholds. It contains no inverse power of $\min \gamma_j$. Each nonunit eigenvalue and each inverse exterior product has the corresponding error $E + \log 2$ per contributing direction. For $\delta \geq m$ and $h = 0$, the model exponent list is $\delta - m$ zeros, $\delta - m$ copies of t_+ , and $\min(t_+, \kappa_j)$, $1 \leq j \leq m$. Sort that list to obtain the individual and exterior statements. The unit factors from $\gamma_j = 1$ remain zero exponents.

PR5. Coordinate transport, mass, and intersecting kernels

For a fixed coefficient isomorphism S , use $T' = STS^{-1}$, $G' = S^{-*}GS^{-1}$, $\Lambda' = \Lambda S^{-1}$. Then $L' = SL$, $H' = SHS^{-1}$, and $\mathcal{Y}'(z) = \mathcal{Y}(z)$ exactly. The original $S = k/2 + iy$ coefficient change and the full-unit isometries must be applied by this paired transport.

Under a common source-mass change $G \rightarrow aG$, $Q_B \rightarrow aQ_B$ and $H_K \rightarrow aH_K$, whereas L, H, A_K and mathscrY are unchanged. The two determinants involving H_K in PR7 acquire the same factor a^m . This verifies the balancing; no actual source mass is changed.

For an additional original onto observation $\text{Xi}:B \rightarrow C$ define $Q_C = (\text{Xi } Q_B^{-1} \text{Xi})^{-1}$ and $L_{\text{Xi}} = Q_B^{-1} \text{Xi } Q_C$. Direct multiplication proves

$$L_{\Xi\Lambda} = L_{\Lambda} L_{\Xi}, \quad \mathcal{Y}_{\Xi\Lambda}(z) = \Xi \mathcal{Y}_{\Lambda}(z) L_{\Xi}. \quad (\text{PR15a})$$

Thus the exact minimum and the full measured resolvent pass through nested observations using the actual maps. Applying PR7 on the composite keeps its full kernel, including any newly formed intersection with $\ker T$. No equality between powers of compressed arithmetic operators is implied by this identity.

At $h > 0$ the positive determinant of A_K refers to the full attained metric on $K/(K \cap V)$. In a fixed frame $[I_Z, I_U]$ of K , its domain Gram is $H_{UU} H_{UZ} H_{ZZ}^{-1} H_{ZU}$ and its image Gram is $(T I_U)^* G(T I_U)$. Their determinant ratio is $\det^+ A_K$. The original full determinant of K additionally contains its inherited determinant on Z . This is the precise remaining factor at an observation/word-kernel intersection.

PR6. Exact complex illustration

In Euclidean $E = \mathbb{C}^3$ use

$$T = \text{diag}(0, 2, 3i), \quad \Lambda = \begin{pmatrix} -1 & 1 & 0 \\ -i & 0 & 1 \end{pmatrix}, \quad K = \mathbb{C}(1, 1, i)^T.$$

Then

$$Q_B = \frac{1}{3} \begin{pmatrix} 2 & i \\ -i & 2 \end{pmatrix}, \quad \mathcal{Y}(z) = \begin{pmatrix} (3z+4)/(3(z+4)) & -4i/(3(z+4)) \\ 3i/(z+9) & (z+3)/(z+9) \end{pmatrix},$$

$$\boxed{\det \mathcal{Y}(z) = \frac{z(3z+13)}{3(z+4)(z+9)}}. \quad (\text{PR16})$$

Here $\Gamma = 2/3$ and $A_K = 13/3$. The off-diagonal entries are nonzero. In the same Q_B metric the separately compressed arithmetic action $T_B = \Lambda T \Lambda$ instead has resolvent determinant $9z^2/(9z^2+52z+36)$. The two exact rational functions disagree, exhibiting the omitted terms in that substitution on specified maps.

PR17-PR31. A signed kq-scale determinant in the original observation

This is an additive derivation from LONG_WORD_SOURCE.md, equations (4)–(10), (37) and (44), and the pinned DCR26–27 receiver. The source's all-direction outer-minimum estimate and its finite guards retain their supplied written-proof status. No independent proof of the imported equilibrium theorem is claimed. All four source cutoffs and the original full physical unit are unchanged.

PR17. Retained arithmetic data and finite inputs

Keep $k \equiv 1 \pmod{4}$ on the source's proved simple-quartet period domain, and put

$$q = (k+1)^2, \quad d = (k-7)^2, \quad \Delta = q - d = 16k - 48, \quad m = 8k - 16,$$

$$Q_k(y) = D_k(y)O_k(y), \quad T_k = D_k(M), \quad M[p] = [yp], \quad S = k/2 + iy.$$

D_k contains the complete pivot-translated interior box and O_k its entire complement. The same original nonzero conductor pivot proves $K \cap \ker T_k = 0$. In the interior/boundary root ordering the conductor is $[A_I, A_{\text{partial}}]$ with A_I invertible, and every vector of K has the exact form (Fb, b) , $\bar{F} = -A_I^{-1} A_{\text{partial}}$. No condition $\mu_0 \neq 0$ is added.

Let π_{partial} be the original boundary-value map and let

$$Q_N^\partial = (\pi_\partial G_N^{-1} \pi_\partial^*)^{-1}, \quad H_{K,N} = I_K^* G_N I_K,$$

$$\Gamma_N = H_{K,N}^{-1/2} (\pi_\partial I_K)^* Q_N^\partial (\pi_\partial I_K) H_{K,N}^{-1/2}. \quad (\text{PR17})$$

Every Γ_N is positive and at most I . Let V_O evaluate degree-below- Δ polynomials at the boundary roots, and define the single fixed injective matrix

$$J = V_O^{-1} \pi_\partial I_K.$$

The uploaded original source minimum is the coefficient Gram $\text{mathfrak{G}}_N$ of $\min \int |D_k|^2 |p + O_{kh}|^2 d\mu_k$, at degrees $\Delta-1$, Δ , $q+\Delta-1$, $q+\Delta$. Its supplied all-direction estimate is

$$e^{c_N - \epsilon_k} I \preceq \mathfrak{G}_N \preceq e^{c_N + \epsilon_k} I, \quad \epsilon_k = o(q),$$

$$c_L = 2q \log q + \eta_L q, \quad c_H = 2q \log q + \eta_H q, \quad \eta_L = 2 \log(4/\pi) - 2, \quad \eta_L - \eta_H = C_\partial/2. \quad (\text{PR18})$$

The low type is $N = q-1, q$; the high type is $N = 2q-1, 2q$. The original image Gram is exactly

$$H_{TK,N} = I_K^* T_k^* G_N T_k I_K = J^* \mathfrak{G}_N J. \quad (\text{PR19})$$

In particular the fixed coefficient factor $\det(J^* J)$ cancels only after the four original signs are applied.

The boundary comparison from `LONG_WORD_SOURCE` (9) is

$$a_k I \preceq Q_N^\partial \preceq b_k I, \quad \omega_k = \log(b_k/a_k) = o(q), \quad (\text{PR20})$$

where, retaining the original arithmetic/Gamma factors ℓ_k, u_k ,

$$a_k = \ell_k / A_{2q}^\partial, \quad b_k = u_k B_k^\partial,$$

$$A_{2q}^\partial = \frac{\Delta e^2}{M_\sigma} (2q+1)^{3/2} (4q+1)^{R_k+1},$$

$$B_k^\partial = \Delta M_\sigma \left(\frac{2\Delta + R_k}{2\delta} \right)^{2(\Delta-1)},$$

$$R_k = k \sqrt{\delta^2 + \gamma^2}, \quad M_\sigma = \sqrt{2\pi}.$$

For explicit spectral constants set

$$V_+ = \Delta \max(1, R_k)^{\Delta-1}, \quad V_- = \Delta \left(\frac{1 + R_k}{2\delta} \right)^{\Delta-1},$$

$$E_k = \epsilon_k + \max\{\log^+(b_k V_+^2), \log^+(V_-^2/a_k)\} = o(q). \quad (\text{PR21})$$

The boundary Vandermonde matrix and its full Lagrange inverse satisfy $\|V_- O\| \leq V_+$, $\|V_- O^{-1}\| \leq V_-$. The exact positive word matrix is $(Q_N^{\text{partial}})^{-1/2} V_- O^{-1} \mathfrak{G}_N V_- O^{-1} (Q_N^{\text{partial}})^{-1/2}$. Consequently every positive word eigenvalue obeys

$$e^{c_N - E_k} \leq \lambda_{j,N} \leq e^{c_N + E_k}, \quad 1 \leq j \leq \Delta. \quad (\text{PR22})$$

PR19 also has relative form bounds $\exp(c_N + E_k) J^* J$. Every zero word mode, exactly d in number, remains outside this positive list.

PR18. Complete observed resolvent spectrum with no angle margin in the error

Let

$$\mathcal{V}_N(z) = \Lambda z(zI + T_k^{\dagger G_N} T_k)^{-1} L_N, \quad L_N = G_N^{-1} \Lambda^* Q_{B,N}.$$

Its determinant is in the original observed value space; the Q_B isometry of PR1 merely gives an orthonormal representation.

Order the actual positive Γ_N eigenvalues $\gamma_{\{j,N\}}$, and put $\kappa_{\{j,N\}} = -q^{-1} \log \gamma_{\{j,N\}} \geq 0$. Define one common regularizer at all cutoffs,

$$z_k(b) = q^{2q} e^{bq}, \quad t_N(b) = \eta_{L/H} - b. \quad (\text{PR23})$$

PR15 gives the complete finite scalar formula

$$\left| \log \det \mathcal{V}_N(z_k(b)) + q \left[(\Delta - m)(t_N)_+ + \sum_{j=1}^m \min((t_N)_+, \kappa_{j,N}) \right] \right| \leq B_k, \quad B_k = \Delta(E_k + \log 2). \quad (\text{PR24})$$

It is uniform for all real b and includes equality at either edge. The operator comparison PR13 also determines each logarithmic eigenvalue with error $E_k + \log 2$, and every inverse exterior product with error at most $\min(p, \Delta)(E_k + \log 2)$. No nonzero angle is replaced by a mean. No inverse power of $\gamma_{\min}$ occurs.

For $d \geq m$, an exact model exponent list consists of $d-m$ zeros, $\Delta-m$ copies of $(t_N)_+$, and $\min((t_N)_+, \kappa_{\{j,N\}})$, $j=1, \dots, m$. Sort it to apply the individual and exterior bounds. At the finitely smaller dimensions PR14, rather than a negative unit multiplicity, supplies the same determinant statement.

PR19. A negative kq -scale determinant throughout the original strict interval

Let $\eta_H < b_0$. At both low endpoints $t_N = \tau$, and at both high endpoints $(t_N)_+ = 0$. Formula PR24 and $0 \leq \min(\tau, \kappa) \leq \tau$ prove

$$\boxed{-2\Delta q\tau - 4B_k \leq \mathcal{R} \log \det \mathcal{Y}_N(z_k(b)) \leq -2(\Delta - m)q\tau + 4B_k.} \quad (\text{PR25})$$

In particular the original observation's resolvent determinant has an eventually negative return with an order- kq margin. At $b_{\text{mid}}=(\eta_L+\eta_H)/2$ the normalized bounds are

$$-8C_\partial \leq \liminf \frac{\mathcal{R} \log \det \mathcal{Y}_N(z_k(b_{\text{mid}}))}{kq} \leq \limsup \frac{\mathcal{R} \log \det \mathcal{Y}_N(z_k(b_{\text{mid}}))}{kq} \leq -4C_\partial. \quad (\text{PR26})$$

This is a sign and a quantitative interval; it is not an asserted equality between the two endpoints. It uses the full observed resolvent, not the resolvent of $(\text{Lambda T L})^\dagger(\text{Lambda T L})$.

PR20. Extend the sign below the high threshold with the original metric order

At $l_i=q-1+i$, $u_i=2q-1+i$ the canonical source inclusions imply $H_{K,u_i} \leq H_{K,l_i}$. Set $B_N=(\pi_{\text{partial } I_K} * Q_N)^{\text{partial } (\pi_{\text{partial } I_K})}$. PR20 implies $B_{u_i} \geq \exp(-\omega_k)B_{l_i}$. Generalized Rayleigh min-max therefore gives

$$\gamma_{j,u_i} \geq e^{-\omega_k} \gamma_{j,l_i}, \quad \kappa_{j,u_i} \leq \kappa_{j,l_i} + \omega_k/q, \quad (\text{PR27})$$

with corresponding sorted indices. This compares both members of the same metric pair, not an assumed monotonicity of angle matrices in fixed Euclidean coordinates.

Define the evaluated nonnegative function

$$f(b) = (\eta_L - b)_+ - (\eta_H - b)_+ = \min\{(\eta_L - b)_+, C_\partial/2\}.$$

The scalar clipping is increasing and 1-Lipschitz in κ . PR27 and $t_l \geq t_u$ show that the sum of the clipped low angles exceeds the sum of clipped high angles by at least $-2m \omega_k/q$. Substitution in PR24 proves

$$\boxed{\mathcal{R} \log \det \mathcal{Y}_N(z_k(b)) \leq -2(\Delta - m)qf(b) + 2m\omega_k + 4B_k.} \quad (\text{PR28})$$

The exact finite negative region is

$$f(b) > \epsilon_k^{\text{edge}} := \frac{m\omega_k + 2\Delta(E_k + \log 2)}{(\Delta - m)q}. \quad (\text{PR28a})$$

Every constant in this guard is supplied in PR18-PR21, and $\epsilon_k^{\text{edge}} = o(1)$. It therefore controls moving regularizers without asserting a step value inside an uncomputed boundary layer.

The finite bound holds for every real b . For each fixed $b < \eta_L$ its leading term is strictly negative; for $b \leq \eta_H$ it saturates at $-(\Delta-m)C_{\text{partial } q}$. For each fixed $b > \eta_L$, PR11 gives absolute convergence of the return to zero. At $b = \eta_L$, PR24 proves only the normalized statement $o(kq)$, without assigning a raw transition value.

PR21. Deep regularization recovers the full original kernel with its exact offset

A completely explicit, sharper angle margin is obtained from the original lattice and graph (Fb, b) . For an upper root ω_{α} put $r_{\alpha, \beta} = \max(|a_{\alpha} - a_{\beta}|, |b_{\alpha} - b_{\beta}|)$. Every distance is at least $2\delta r_{\alpha, \beta}$ and at most $8j$ grid points occur at shell j . Thus

$$\sum_{\beta \neq \alpha} \log(k/r_{\alpha\beta}) \leq \sum_{j=1}^k 8j(k/j - 1) = 4k(k-1) \leq 4q.$$

The complete Lagrange denominator is at least $(2\delta k)^{q-1} \exp(-4q)$. Successive Gamma multiplication bounds retain every numerator root and give norm at most $\sqrt{M_{\sigma}}(2q + R_k)^{q-1}$. Summing the squared norms of all q Lagrange columns proves

$$G_N^{val} \preceq u_k B_{lat,k} I, \quad B_{lat,k} = q M_{\sigma} e^{8q} \left(\frac{2q + R_k}{2\delta k} \right)^{2(q-1)}. \quad (\text{PR29a})$$

Every lifting polynomial has degree $q-1$, so it is admitted at all four original cut-offs. This strengthens the first note's cruder B_{full} by retaining the complete lattice-distance product rather than charging the smallest distance $q-1$ times.

The exact strip map $N_k = [R, I]$ satisfies $N_k M_A = 0$. In the dual root order its transpose identifies the actual graph $F = R^T$; hence $\|F\| \leq \sqrt{q} C_{\text{path}}^{10k}$, where $C_{\text{path}} = A_1/|a_*|$ is the retained nonzero pivot ratio. The boundary norm of a kernel vector (Fb, b) is at least $a_k \|b\|^2$, whereas its full norm is at most $u_k B_{lat,k} (1 + q C_{\text{path}}^{20k}) \|b\|^2$. Therefore

$$\Gamma_N \succeq \vartheta_k I, \quad \vartheta_k = \min \left\{ 1, \frac{a_k}{u_k B_{lat,k} (1 + q C_{\text{path}}^{20k})} \right\} > 0. \quad (\text{PR29})$$

All fixed-period and original source constants remain. Since their logarithms here are $o(q)$, the explicit bound has

$$-\log \vartheta_k = 2q \log(q/k) + (8 - 2 \log \delta)q + o(q). \quad (\text{PR29b})$$

This is the asymptotic of the specified conservative lower bound, not a value assigned to any actual angle. Combining it with PR22 evaluates a lower rate for EVERY original-kernel word-energy direction:

$$\liminf_{k \rightarrow \infty} \min_N \left\{ \frac{\log \lambda_{\min}(A_{K,N}) - 2q \log k}{q} - \eta_{L/H} \right\} \geq -8 + 2 \log \delta. \quad (\text{PR29c})$$

The exact finite version is $A_{K,N} \geq \vartheta_k \exp(c_N - E_k)I$. This is the full rectangular energy of T_k on the original K , not the square of a compressed arithmetic endomorphism. Its word degree is $d = q - O(k)$; all its primary zero modes outside this transverse K remain in H .

Choose the same positive regularizer at all four cutoffs,

$$z_k^{\text{deep}} = \vartheta_k e^{c_H - E_k - q}.$$

The positive compression $A_{K,N}$ satisfies $A_{K,N} \geq \vartheta_k \exp(c_H - E_k)I$. Expand the exact PR6 determinant at this regularizer. The two finite errors obey

$0 \leq \log \det(I + z A_{K,N}^{-1}) \leq m \exp(-q)$, $0 \leq \log \det(I + z B_{+,N}^{-1}) \leq \Delta \exp(-q)$.

Since $A_{K,N} = H_{K,N}^{-1/2} H_{TK,N} H_{K,N}^{-1/2}$ and PR19 retains the same J at all endpoints, the four-return is

$$\left| \mathcal{R} \log \det \mathcal{Y}_N(z_k^{deep}) + (\Delta - m)C_{\partial q} + \mathcal{K}_k \right| \leq E_k^{deep}, \quad (\text{PR30})$$

$$\mathcal{K}_k = \mathcal{R} \log \det H_{K,N},$$

$$E_k^{deep} = \min\{4(m + \Delta)E_k, 2m\omega_k + 4(\Delta - m)E_k\} \\ + 2(m + \Delta)e^{-q} = o(kq).$$

For the sharper second term in this error, use the exact PR8a factor with $\rho = \Delta - m$. PR22 gives $\exp(-c_N - E_k)I \leq X_{i,N} \leq \exp(-c_N + E_k)I$. Hence

$$\log \det \Xi_N = -\rho c_N + e_N, \quad |e_N| \leq \rho E_k.$$

Also $\log \det \Gamma_N = \log \det B_N - \log \det H_{K,N}$, where $B_N = (\pi_{\text{partial}} I_K)^* Q_N^{\text{partial}} (\pi_{\text{partial}} I_K)$. The full source minima make B_N decrease with cutoff, and PR20 gives

$$0 \leq \mathcal{R} \log \det B_N \leq 2m\omega_k.$$

Thus PR8a and the same thermal remainder prove the directed bound

$$-4\rho E_k - \varepsilon_{reg} \leq \mathcal{R} \log \det \mathcal{Y}_N(z_k^{deep}) + \rho C_{\partial q} + \mathcal{K}_k \leq 2m\omega_k + 4\rho E_k + \varepsilon_{reg}, \quad \varepsilon_{reg} = 2(m + \Delta)e^{-q}. \quad (\text{PR30a})$$

The complementary space and its full Schur minimum may vary with N ; its form bound is valid on every one of those actual spaces, so no common eigenvectors or generic orientation is needed.

There is now a fixed-exponent common regularizer as well. For any fixed

$$b < \eta_H - 8 + 2 \log \delta, \quad z_k = k^{2q} e^{bq}, \quad (\text{PR30b})$$

PR29c makes $z_k / \min_N \lambda_{\min}(A_{K,N})$ exponentially small. The positive full word modes are still larger, by PR22. Therefore the same leading identity PR30 holds at PR30b with an explicitly exponentially decreasing regularization error. This regularizer is at the $k^{(2q)}$ arithmetic-root scale, rather than specified using an unknown original Gram eigenvalue.

The coefficient is $\Delta - m = 8k - 32$, not Δ or m . It arises from the retained zero-word modes in PR6 and cancels neither the original kernel volume nor the positive word determinant.

The pinned DCR27 receiver states, with its full explicit error $E_k^K = o(kq)$,

$$0 \leq \mathcal{K}_k \leq g_k C_{\text{partial } q} + E_k^K, \quad g_k = \Delta - v.$$

Consequently

$$-(24k - 80 - v)C_{\partial q} - E_k^K - E_k^{deep} \leq \mathcal{R} \log \det \mathcal{Y}_N(z_k^{deep}) \leq -(8k - 32)C_{\partial q} + E_k^{deep}. \quad (\text{PR31})$$

The unknown kernel allocation remains in the exact equality PR30. The new observable has a sign without assigning that unknown value.

PR32-PR44. Activation, all-direction source extraction, and exact sharpness

PR32. The same measured resolvent at the actual source activations

Retain the exact source covariance family

$$G_N(\alpha)^{-1} = (1 - \alpha)G_N^{-1} + \alpha(G_N^J)^{-1}, \quad 0 \leq \alpha \leq 1.$$

The original T_k , Λ_k , I_k , interior polynomial D_k and boundary polynomial O_k are unchanged. Define the observed resolvent through the actual $Q_B(\alpha)$ and its minimum section at each α . Every finite identity PR1-PR16 holds on this family, at both endpoints and with the full normal-source cross Gram.

For $\alpha > 0$ the covariance order gives

$$G_N^J \preceq G_N(\alpha) \preceq \alpha^{-1} G_N^J. \quad (\text{PR32})$$

The supplied complete joint-source comparison is $L_k^{-1} T_{\text{val}} \leq G_N^J \leq B_k T_{\text{val}}$, with $\log(L_k B_k) = o(q)$. In that stated reference T_k is diagonal with values $D_k(\omega)$ at the original boundary roots and zero at the interior roots. Thus

$$\|T_k\|_{G_N(\alpha)}^2 \leq \alpha^{-1} L_k B_k (2k R_h)^{2d} =: M_k(\alpha). \quad (\text{PR33})$$

This uses the entire original scalar/tensor minimum: α is its stated penalty, not a rescaling of the original physical measure. Compression in the actual observation and $\text{rank}(T_k) = \Delta$ now give

$$0 \leq -\log \det \mathcal{Y}_{N,\alpha}(z) \leq \Delta \log \left(1 + \frac{M_k(\alpha)}{z} \right), \quad (\text{PR34})$$

$$\|I - \mathcal{Y}_{N,\alpha}(z)\|_1 \leq \Delta \frac{M_k(\alpha)}{z + M_k(\alpha)}.$$

The trace norm is in the actual observed metric. The rank factor is Δ , not q -m. At $z = z_k(b)$ from PR23, for b in a fixed bounded interval,

$$\log \frac{M_k(\alpha)}{z_k(b)} \leq -\log \alpha - 2q \log(q/k) + O_h(q). \quad (\text{PR35})$$

Consequently, at every fixed $0 \leq \beta < 1$ and $\alpha_k = \exp[-2\beta q \log(q/k)]$, the observed determinant and its four-return tend to zero absolutely. More generally PR35 is the finite criterion on any moving activation; it does not assign an unproved law on the critical scale $\beta = 1$. In particular the canonical negative return of PR25 or PR28 and this activated vanishing concern exactly the same arithmetic word and observation. Their change of metric remains in PR32 and in the complete source determinant identities; it is not a zero-cost substitution.

Uniform control over every joint-source regularizer

For any two positive metrics $aG_0 \leq G_1 \leq bG_0$ on the SAME E , the positive squared singular values of T and of T restricted to the fixed K each change by factors between a/b and b/a . This follows from the two actual numerator/denominator Rayleigh norms; their kernels are unchanged. Since $\log(1 + \lambda/z)$ is 1-Lipschitz in $\log \lambda$, PR8 gives the finite comparison

$$|\log \det \mathcal{Y}_{G_1}(z) - \log \det \mathcal{Y}_{G_0}(z)| \leq (\delta + p) \log(b/a), \quad z > 0, \quad (\text{PR35a})$$

where $p=\text{rank}(T|K)$, with intersections retained exactly as in PR8. For the original joint-source family $p=m$, $\Delta=\Delta$ and the common reference T_{val} is the same at all four cutoffs. Therefore

$$\sup_{z>0} |\mathcal{R} \log \det \mathcal{Y}_{N,J}(z)| \leq 4(\Delta + m) \log(L_k B_k) = o(kq). \quad (\text{PR35b})$$

The powers of $\log z$ cancel under the four signs, so the finite zero-regularizer limit of the return has the same bound. This is a uniform all-regularizer result for the original joint source, including the scale PR30b. It does not compare the canonical source to the joint source at zero cost.

PR33. Full-root innovations from the second supplied note

Use the original lower-root evaluation matrix E_N , the actual invariant rows W_N , and the Gamma source cutoff $D=N-v$. Define

$$C_N = W_N [I - E_N^* (E_N E_N^*)^{-1} E_N] W_N^*.$$

For the original appended columns e_N, w_N , set

$$\kappa_N = 1 + e_N^* (E_N E_N^*)^{-1} e_N, \quad v_N = w_N - W_N E_N^* (E_N E_N^*)^{-1} e_N.$$

The unit new kernel direction is $(-E_N (E_N E_N^*)^{-1} e_N, 1)^T / \sqrt{\kappa_N}$. It is orthogonal to the entire old lower-root ideal, so the exact update is

$$C_{N+1} = C_N + v_N v_N^* / \kappa_N. \quad (\text{PR36})$$

This repeats the second note's formula as the input map to the new estimates. It is not an omitted lower-root projection. Its complex cross term is explicit.

For $i=0,1$, collect all q updates from $l_i=q-1+i$ through $u_i=2q-1+i$. Let $V_i^{\wedge\{\text{all}\}}$ be their innovation column matrix and $\bar{X}_i = C_{l_i}^{-1/2} V_i^{\wedge\{\text{all}\}}$. Then

$$G_i = \log \det(I + X_i X_i^*) \geq 0, \quad \mathcal{R} \log \det C_N = -G_0 - G_1. \quad (\text{PR37})$$

$A_i = [I_r, X_i]$ is a physical orthonormal-column source map in the explicitly normalized target coordinates. Its first r source columns are the isometry $R_{l_i}^{\dagger} C_{l_i}^{-1/2}$ in the old ideal; the appended columns are the actual orthogonal innovations. The old zero-value source summand remains separate.

PR34. Selected original columns control every spectral direction

Choose r actual columns of A_i as a nonsingular matrix V_i . After the exact column permutation write

$$A_i P_i = V_i [I_r, T_i], \quad |(T_i)_{ab}| \leq 2. \quad (\text{PR38})$$

There are q unselected columns. The exact source covariance is

$$A_i A_i^* = V_i (I + T_i T_i^*) V_i^*.$$

Because $\text{Tr}(T_i T_i^*) \leq 4rq$,

$$V_i V_i^* \preceq A_i A_i^* \preceq (1 + 4rq) V_i V_i^*. \quad (\text{PR39})$$

Thus all corresponding ordered eigenvalues obey

$$0 \leq \log \lambda_j(A_i A_i^*) - 2 \log s_j(V_i) \leq \log(1 + 4rq), \quad 1 \leq j \leq r. \quad (\text{PR40})$$

Every p-dimensional exterior volume has error at most $p \log(1+4rq)$. Every fixed restriction or full attained quotient of the inverse metric has the same relative form bound and rank-p logarithmic error. This is a full spectral statement, not just the determinant estimate of the supplied note.

The sharper full determinant bound remains

$$G_i = \log |\det V_i|^2 + \rho_i, \quad 0 \leq \rho_i = \log \det(I + T_i T_i^*) \leq r \log(1 + 4q). \quad (\text{PR41})$$

Its inverse, including all phase data, is exactly

$$(A_i A_i^*)^{-1} = V_i^{-*} (I + T_i T_i^*)^{-1} V_i^{-1}.$$

The residual has eigenvalues between $1/(1+4rq)$ and 1. Only after the displayed coordinate map V_i^{-1} has been applied is that a bounded-coefficient remainder. An original action and marked class must also transform to $V_i^{-1} A V_i$ and $V_i^{-1} x$. The residual cannot be dropped in a complex-current calculation.

PR35. The number of swaps improves to $O(kq)$

Retain the scalar Gamma/Hankel expression $A_{\{k,v\}}$ and full finite error E_k^{src} from INNOVATION_AND_ES_SOURCE, equations (8)-(9). They state

$$|\mathcal{K}_k - \log \mathcal{A}_{k,v} + G_0 + G_1| \leq E_k^{\text{src}}. \quad (\text{PR42})$$

For reference its unchanged scalar is

$$\mathcal{A}_{k,v} = \frac{\Delta_{g-1}(q') \Delta_g(q') \Delta_{q-1}(q-v) \Delta_q(q-v)}{\Delta_{q+g-1}(q') \Delta_{q+g}(q') \Delta_0(q-v)},$$

$$\Delta_j(a) = \det[\nu_{2a+i+l}]_{i,l=0}^j, \quad \nu_{2b} = (2b)! [t^{2b}] (\cos t)^{-1/2}, \quad \nu_{2b+1} = 0.$$

The original Gamma mass cancels because numerator and denominator contain equal total matrix dimensions. E_k^{src} is the exact supplied full-source error, $O(q \log(q+2))$ on its retained finite guards.

Monotonicity of the original kernel metric gives $\text{mathcalK}_k \geq 0$, so PR42 proves

$$G_0 + G_1 \leq \log \mathcal{A}_{k,v} + E_k^{\text{src}}.$$

The DCR26 receiver and CGS17 unprojected determinant bound imply, by subtracting the two equations for the SAME mathcalK_k ,

$$|\log \mathcal{A}_{k,v} - gC_{\partial q}| \leq E_k^{\text{src}} + B_{DCR,k} + \sum_N K_N = o(kq).$$

No value for mathcalK_k was needed to obtain this scalar estimate. Starting with the identity columns, each accepted swap whose certified modulus exceeds $3/2$ raises selected squared volume by more than $(3/2)^2$. The total selected volume is bounded above by $\exp(G_0 + G_1)$. Hence the total number of swaps in BOTH searches is at most

$$\boxed{\frac{\max(0, \log \mathcal{A}_{k,v} + E_k^{\text{src}})}{2 \log(3/2)} = O(kq)}. \quad (\text{PR43})$$

This improves the older $O(kq \log q)$ endpoint bound. It bounds swaps, not the bit cost of obtaining reliable values of the original period-dependent entries. The supplied overlapping acceptance tests ≤ 2 and $> 3/2$ remain valid.

Let $Z_k^{\text{piv}} = \log A_{\{k,v\}} - \log |\det V_0 \det V_1|^2$. PR42 and PR41 give the supplied kernel interval with every residual retained. Together with PR30 the NEW measured resolvent interval is

$$\begin{aligned} -(\Delta - m)C_{\partial q} - Z_k^{\text{piv}} - E_k^{\text{src}} - E_k^{\text{deep}} &\leq \mathcal{R} \log \det \mathcal{Y}_N(z_k^{\text{deep}}) \\ &\leq -(\Delta - m)C_{\partial q} - Z_k^{\text{piv}} + E_k^{\text{src}} + 2r \log(1 + 4q) + E_k^{\text{deep}}. \end{aligned} \quad (\text{PR44})$$

Both selected determinants consist of actual original source columns. The exact values of those columns have not been numerically supplied or evaluated here.

Exact sharpness with the same finite word and observation

This is a declared auxiliary source family, not an arithmetic xi packet. It checks that the word spectrum and boundary quotient alone do not fix the kernel allocation. On C^Δ direct-sum C^Δ retain

$T = \text{diag}(0, I_\Delta)$, $K = \text{span}\{(e_j, e_j) : 1 \leq j \leq m\}$,

and the same onto observation with exactly that kernel. For $a \geq 1$ use either

$$G_\pm(a) = \begin{pmatrix} aI & \pm(a-1)I \\ \pm(a-1)I & aI \end{pmatrix}.$$

Their common boundary quotient is $((2a-1)/a)I$. Their complete positive word spectrum is Δ copies of $a^2/(2a-1)$. Every V-zero mode is retained. The two kernel Grams and compressed kernel energies are instead

$$H_{K,+} = (4a-2)I_m, \quad A_{K,+} = a/(4a-2)I_m,$$

$$H_{K,-} = 2I_m, \quad A_{K,-} = a/2 I_m.$$

Therefore their exact measured determinants are

$$\det \mathcal{Y}_\pm(z) = \frac{z^{\Delta-m}(z + A_{K,\pm}^{\text{scalar}})^m}{(z + a^2/(2a-1))^\Delta}.$$

Both metrics increase with a in positive-form order, so setting $a_{\text{low}} = R^2$, $a_{\text{high}} = R$ gives nested positive quotient metrics. They have an actual finite source realization: take a square-root covariance source at the low endpoint and add a square root of the positive covariance difference at the high one. At $z = R^{3/2}$, with two copies of each endpoint,

$$\mathcal{R} \log \det \mathcal{Y}_+ = -\Delta \log R + O(1), \quad \mathcal{R} \log \det \mathcal{Y}_- = -(\Delta - m) \log R + O(1).$$

Meanwhile their kernel returns are $2m \log R + O(1/R)$ and zero. These exact maps attain both rank bounds in the strict-interval argument. They demonstrate why the actual invariant rows still have to be evaluated for an equality value. The source and metric choices of the original programme are not replaced by these illustrations.

Sources and verification scope

LONG_WORD_SOURCE.md is the first upload, preserved byte for byte. Its equation (8) supplies the all-direction image form; (9) the boundary quotient; (10) the positive word spectrum; (37) the original kernel angle; (44) its finite margin.

INNOVATION_AND_ES_SOURCE.md is the second upload, preserved byte for byte. This continuation uses its RH original-root formulas (1)–(10). Its later ES receiving bounds stay in their stated label/receiving metrics. No new ES-to-RH map or counterexample claim is introduced here.

DCR1-27 was inspected at GitHub commit `bee43be41d2f6f872869d5e258fa5f752335a5ad`, path `workbenches/splitzero-tandem/continuations/20260921-local-covariance-activation/009/CONDUCTOR_DIVISOR_COVARIANCE_RETURN.tex`, blob `256bfb0f4eb295bf3094c81f6c166cafdb96c316`. Its EIQ/OOQ provider remains an inherited input at the source's written-proof status, not a theorem newly reproved here.

The bounded interpolation-factor context is classical: M. Gu and S. C. Eisenstat, Efficient Algorithms for Computing a Strong Rank-Revealing QR Factorization, SIAM Journal on Scientific Computing 17 (1996), 848–869, doi:10.1137/0917055. PR39–43 were derived directly for the specified source columns. The Gamma convention is NIST DLMF 18.23.7, with its source note crediting Ismail (2009).

The checker uses exact Gaussian-rational auxiliary matrices. It tests the full resolvent determinant, the retained intersection zeros, noncommuting metric blocks, all principal-minor form bounds in the fixtures, full-root innovations, selected-column residuals and the exact sharpness examples. These checks do not validate an unspecified native period, xi packet, or the imported analytic asymptotics.

Part VI

Human sources and exact programme receivers

Human sources and exact programme receivers

The complete mathematical arguments in this reader retain every original source form and its stated analytic input status. Numerical fixtures do not establish a native asymptotic coefficient. Earlier proofs are linked to fixed public editions below, and their complete programme sources accompany this edition.

Human sources used in the new proofs

- Geneviève Dusson, Israel Michael Sigal and Benjamin Stamm, *The Feshbach-Schur map and perturbation theory*, arXiv:2105.02058v1, original-author TeX AriFestFSM_arxiv.tex, theorem thm:isospF, equations Fesh, QP, Hlam-deco, Ulam-def, source lines 398–489. RM1–18 applies the construction with the original minimum-section projection and the full positive word energy. The block elimination, smallest contributing space and reconstruction are proved in the reader. The original archive is retained locally with its exact identity in the reading ledger.
- R. A. Askey and R. Roy, NIST DLMF Chapter 5, Gamma product 5.8.3, original equation TeX. CS11 uses it for the original physical Gamma lower bound. The full coefficient and integral argument is supplied there.
- T. H. Koornwinder, R. Wong, R. Koekoek and R. F. Swarttouw, with W. P. Reinhardt, NIST DLMF Chapter 18: recurrence 18.22.8, generating function 18.23.7, Legendre generating function 18.12.11. Their retained original equation TeX supplies the conventions used in CS4, CS10–11 and the inherited AS estimates. The physical source mass is kept in every use.
- J. L. W. V. Jensen, *Sur un nouvel et important théorème de la théorie des fonctions*, Acta Mathematica 22 (1899), 359–364, DOI10.1007/BF02417878. The inherited AS source identifies the zero-count formula; the finite disk count and required quotient estimates are proved in the retained programme source. This edition does not claim a new reading of the complete historical paper.
- P. G. Lejeune-Dirichlet, the 1837 arithmetic-progression paper in Ralf Stephan’s English translation, arXiv:0808.1408v2. ES17 uses the opening coprime-progression theorem for its exactly checked progression. The retained TeX was read at lines 1–130, including the translator credit. The translation is distinguished from original German typesetting; the complete historical proof is not claimed newly read or reproved.

Exact public programme proofs

- AS1–64: every original projection direction.
- FW1–49: full word, literal roots, and measured heat.
- RX1–20: exact phased source innovations.

- AD1-14: directed whole-window angle spectrum.
- FI1-17 and FR1-4: finite rational original kernel receiver.
- CI1-26 and CL1-18: full cutoff and invariant covariance maps.
- IC1-42: complete unpaired-root and collision proof.

The primary-word determinant calculation PR1-44 is the supplied complete proof, retained as a separate part. Its finite identities use the original positive metric and observation. Its arithmetic asymptotics retain the existing DCR/OOQ/EIQ source status and finite guards; no finite checker upgrades those analytic inputs. The independent RM derivation proves the complete kernel correction and full-source error theorem. CS strengthens the incoming complete-row sampling result by paying for root-coefficient and final matrix errors. The RG proof records its radius repair explicitly. The ES calculation is an original-label receiver computation and does not identify that four-dimensional receiver with the growing RH observation.

The human attribution supplied with the ES arrivals identifies Levent Alpöge's originating announcement, crediting Akhil Mathew and Claude Fable 5, and separately credits Terence Tao's exposition. That attribution is retained as source provenance, not represented as a newly read announcement or a Jacobian identification. The explicit four-label linear receiver and its complete proofs are specified in ES1-37.